

Mbarara Integrated Concrete Products Factory

A4 portrait consultancy report

Currency basis: UGX only

Commercial planning horizon: 2026-2035

Prepared: June 2026



Document Control

This PDF is generated from a reusable ReportLab generator and presents a pre-feasibility investment case for a concrete products factory serving Mbarara, western Uganda and selected East African corridors.

All monetary values are stated in UGX only. Values are indicative planning assumptions, not quotations. Final investment decisions should be supported by supplier quotations, professional engineering, tax advice, legal review, geotechnical work and customer due diligence.

Field	Record
Project	Mbarara Integrated Concrete Products Factory
Location basis	Mbarara City / western Uganda industrial catchment
Report form	A4 portrait investor-grade PDF report
Planning horizon	2026-2035
Financial basis	UGX nominal, before financing structure unless noted
Prepared	June 2026

Investor use: screening, bankability discussion, board preparation, lender questions and procurement planning.

Important Notice and Limitations

This report is intentionally investor-ready in structure but remains a pre-feasibility model. It uses public-source market context and internally generated operating assumptions for product mix, throughput, CAPEX, OPEX and 10-year forecast.

The model should be refreshed before financial close with binding supplier quotes, local construction BoQs, electricity connection studies, water supply confirmation, tax advice, product certification requirements and signed customer off-take indications.

Limitation	Implication	Recommended diligence action
Market sizing	Demand is inferred from construction, urbanization and infrastructure drivers.	Run contractor interviews and collect public tender pipeline data.
CAPEX	Indicative equipment and civil costs can move materially.	Obtain three supplier quotations and a local quantity surveyor estimate.
Regulatory	Approval route may vary with site and environmental sensitivity.	Screen site with Mbarara City, NEMA and relevant lead agencies.
Tax	Model includes generic tax assumptions.	Confirm incentives, VAT treatment and depreciation allowances with URA/UIA advisors.

Table of Contents

Chapter and major-section page numbers are generated automatically by the PDF builder.

1. Executive Summary	6
2. Market Analysis	11
Demand Drivers	12
3. Competitor Analysis	20
Competitor Landscape	21
4. Product Portfolio	30
Portfolio Master Table	31
5. Mix Ratios	47
Mix Ratio Summary	48
6. Raw Materials	55
Raw Material Control Register	56
7. Machinery Options	66
Machinery Option Comparison	67
8. Factory Layout	77
Indicative Factory Layout	78
9. Electricity and Water Consumption	86
Utility Demand Model	87
10. Laboratory and Quality Control	95
Quality Control System	96
11. Legal and Regulatory Offices	106
Regulatory Office Map	107
12. CAPEX and OPEX	119
CAPEX Schedule	120

13. 10-Year Financial Forecast	131
Forecast Summary Table	132
14. Risk Register	144
Risk Register Master Table	145
15. Implementation Roadmap 2026-2035	154
Roadmap Overview	155
16. East African Expansion Plan	167
Regional Expansion Logic	168
17. Appendices	176
Appendix A: Assumption Register	177

1. Executive Summary

Recommended base case: a staged integrated factory with automatic block/paver production, dedicated kerb/channel/culvert capability and a ready-mix plant introduced after market traction is proven.

FUNDING ENVELOPE

UGX 36.13 bn

includes contingency and opening working capital

OPERATING BREAK-EVEN

Year 3

base case reaches positive operating leverage after ramp-up

TARGET MARKETS

Western Uganda

Mbarara, Bushenyi, Ntungamo, Kabale, Isingiro and cross-border corridors

The investment thesis is to establish a quality-led, mechanized concrete products factory in Mbarara that can serve residential, institutional, road, drainage and commercial concrete demand with reliable supply, certified quality and disciplined dispatch.

1 Investor lens

What the chapter means for return, risk and bankability.

2 Operating lens

Factory routines, people, quality and throughput implications.

3 Decision lens

Actions required before financial close and procurement.

Executive Dashboard

The factory is positioned as a formal, certified and reliable alternative to fragmented manual production. The model rewards scale, disciplined curing, reliable delivery and a portfolio that balances daily building materials with infrastructure-oriented products.

The base case assumes a 2026 setup year, commercial launch in 2027 and progressive ramp to regional distribution by 2035.



Investment Rationale

Mbarara is a strategic production base because it combines western Uganda demand, transport access to construction corridors and proximity to Rwanda, DRC-facing routes and fast-growing secondary towns.

The investor case is not simply to make blocks. The value proposition is a formal building materials platform: documented mix designs, cube testing, uniform dimensions, reliable curing, scheduled deliveries, branded stock availability and contractor credit discipline.

The recommended approach is staged. Commission the core block and paver line first, stabilize quality and cash conversion, then add ready-mix and heavier civil products as anchor contracts and dispatch capability mature.

1 Scale

Mechanized output lowers unit cost versus manual yards once utilization rises.

2 Quality

Lab-backed batches help win institutional and contractor confidence.

3 Portfolio

Blocks drive frequency; pavers and civils raise margin depth.

4 Location

Mbarara supports western Uganda and cross-border expansion options.

Recommended Strategic Configuration

The base case uses a fully automatic vibropress production line with modular molds, covered curing, a basic but credible concrete laboratory, yard handling equipment and a sales/distribution model anchored by contractor relationships.

Ready-mix concrete should be planned into the site from day one but commercially scaled once the factory has proven dispatch discipline, quality records, water control and fleet management.

Decision area	Recommendation	Reason
Production line	Automatic vibropress core line	Improves dimensional consistency, production speed and traceability.
Curing	Covered wet curing plus controlled chambers	Protects strength development and minimizes rejected batches.
RMC	Install in phase 2 or late phase 1	Higher working-capital and fleet demands require disciplined dispatch.
Sales model	Contractor, municipal and dealer channels	Balances volume orders with repeat trade and route coverage.
Quality	UNBS/NBRB-oriented QC system	Differentiates against informal supply and supports institutional tenders.

Top Ten Investor Questions

The questions below should be answered before financial close. They convert the report from pre-feasibility into a lender-ready investment memorandum.

No.	Question	Evidence required
1	What site will be secured?	Title/lease, zoning confirmation, access, drainage, geotechnical and utilities route.
2	Which equipment supplier is bankable?	Technical offer, warranty, local support, mold list, spares and commissioning plan.
3	Who are anchor customers?	Letters of intent from contractors, estates, road contractors and institutional buyers.
4	How will quality be certified?	UNBS product route, lab plan, cube testing protocol and retained sample register.
5	What is the working capital cycle?	Credit policy, receivable aging assumptions, stock turns and customer payment terms.
6	How resilient is power and water?	UEDCL connection assessment, standby generation, NWSC/borehole plan and recycling.
7	Can price survive informal competition?	Delivered-cost analysis, contractor interviews and certification premium testing.
8	What approvals are gating?	NEMA screening, city permits, URSB/URA/NSSF, OHS and trade license calendar.
9	How will management control shrinkage?	Weighbridge, ERP, batch tickets, dispatch reconciliations and stock takes.
10	When should expansion start?	Only after service level, receivable days and quality pass rate remain within thresholds.

2. Market Analysis

Public data from the World Bank and UBOS points to a large, young and growing economy with industry, manufacturing and construction as important growth channels. The factory should treat the market as a corridor system: urban Mbarara plus western districts, road projects, institutions and selected export-adjacent routes.

UGANDA POPULATION

45.9m

UBOS 2024 Census portal total population

GROWTH CONTEXT

6%+ GDP

World Bank reported strong growth into 2025/2026

MARKET FORM

Fragmented

formal producers, local yards, dealers and site casting compete

Uganda’s economy and construction demand provide a supportive context for local building-material production, while Mbarara gives the project a regional distribution base rather than a single-city market.

1 Investor lens

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Demand Drivers

The most important demand drivers are urban housing, estate development, schools and clinics, road drainage, parking and paving, commercial slabs, small industrial sheds, and public-sector works.

Concrete products are heavy and low value per kilogram. This gives a local factory a delivered-cost advantage if it can maintain quality and avoid unreliable lead times.

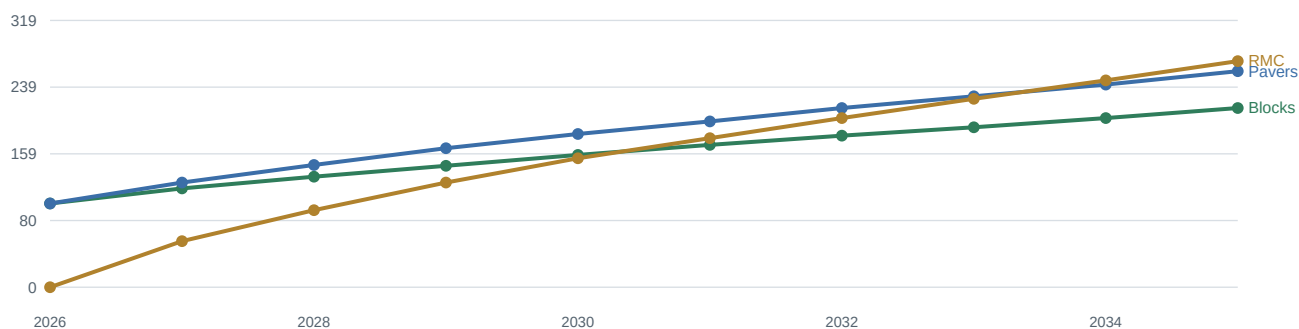
Demand driver	Products pulled	Commercial implication
Urban housing	4-inch, 6-inch and 8-inch blocks	High-frequency demand; dealer and contractor relationships matter.
Commercial estates	blocks, pavers, ready-mix	Requires credit control, delivery planning and consistent dimensions.
Municipal roads	kerbs, pavers, drainage channels	Needs tender documents, quality files and delivery scheduling.
Rural access roads	culverts, channels, concrete	Batch traceability and reinforcement compliance are key.
Tourism and hospitality	cobblestone, hexagonal and permeable pavers	Premium visual products can raise margin.

Macro Context

The World Bank describes Uganda as a young, growing country and reported broad-based economic performance with industry, manufacturing and construction among strong areas. This does not automatically translate into factory sales, but it supports a positive screening view.

UBOS maintains annual GDP, quarterly GDP and construction input price publications. These should become the monthly reference set for management once the factory is operational.

Indicative demand index by segment, 2026-2035 (2026 = 100)



Western Uganda Catchment

Mbarara should be treated as the operating hub for a wider catchment rather than the only source of demand. The immediate ring includes Mbarara City, Isingiro, Kiruhura, Ibanda and Bushenyi; the second ring extends to Ntungamo, Kabale, Rukungiri, Kasese and Fort Portal depending on transport economics.

The factory should price by delivered zone, not one ex-works price. Blocks and kerbs are freight-sensitive; RMC is time-sensitive; pavers can travel farther if packed well and loaded efficiently.

Zone	Target radius	Best-fit products	Route logic
Core Mbarara	0-25 km	blocks, pavers, RMC	same-day dispatch and repeat contractors
Western ring	25-100 km	blocks, kerbs, pavers	scheduled dealer drops and project deliveries
Highland route	100-180 km	pavers, culverts	higher-value loads and less frequent drops
Cross-border prep	route-based	pavers, culverts	depot or partner model after domestic proof

Customer Segmentation

The commercial plan should separate cash buyers, contractors, institutions, public projects, dealers and ready-mix customers. Each segment has different margin, credit and service requirements.

Management should avoid giving contractor credit before the plant has a live receivables dashboard and signed delivery acceptance protocol.

Segment	Buying behavior	Service requirement	Credit stance
Small builders	frequent small lots	visible stock and loading speed	cash or mobile payment
Contractors	bulk orders by project phase	delivery accuracy and documents	approved limits only
Institutions	planned procurement	quality files and tax compliance	formal payment cycle
Dealers	replenishment loads	consistent margins and packaging	secured rotation
RMC customers	day-specific pours	slump control and delivery tickets	deposit and pour sign-off

Pricing Architecture

The factory should publish ex-works list prices, delivered zone prices and project quotation rules. Discounting should be tied to volume, payment speed and pallet return discipline.

A basic price discipline rule is that any discount below standard gross margin must be approved by finance with a reason code.

Price layer	Purpose	Control
Ex-works list	Transparent baseline for dealers and contractors	Reviewed monthly against cement and fuel.
Delivered zone	Captures freight and loading costs	Distance bands with minimum load sizes.
Project quote	Supports tenders and large jobs	Validity period and escalation clause.
Cash discount	Improves liquidity	Only for prompt payment or advance deposits.

Procurement Pipeline Targets

The first 18 months should prioritize conversion of a practical pipeline rather than generic brand awareness. The best early customers are those that value reliable dimensions, delivery and documentation.

The sales team should maintain a rolling pipeline by project stage: identified, qualified, quoted, negotiated, won, delivered and collected.

Pipeline source	Products	Evidence to collect
Residential estates	blocks, pavers	project size, build stages, creditworthiness
Schools and clinics	blocks, channels, pavers	procurement calendar and technical requirements
Road contractors	kerbs, channels, culverts	BOQ, specification and delivery windows
Commercial slabs	RMC	pour schedule, pump needs and site access

Market Data to Refresh Monthly

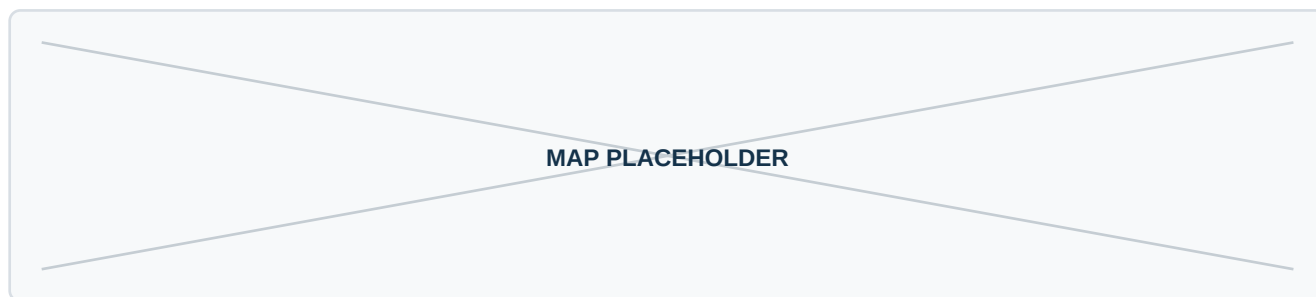
Investor-grade operation requires a market dashboard. The factory should not rely on annual reports alone; it should track leading indicators that move product demand and margin.

The initial dashboard should include cement landed price, fuel price, aggregate cost, construction input index, tender pipeline, receivable days and competitive delivered prices.

Indicator	Source or owner	Decision supported
Cement landed price	procurement	pricing and gross margin
Fuel and transport	logistics	delivered-zone pricing
Construction input index	UBOS publications	escalation clauses and budget review
Tender pipeline	sales manager	order planning and inventory
Receivable days	finance	credit limits and cash conversion

Market Visual Placeholder

The final investment memorandum should insert a GIS-style map showing Mbarara, production site, delivery rings, industrial parks, quarry sources, cement supply routes and target cross-border routes.



Recommended insert: catchment and logistics map with 25 km, 100 km and 180 km delivery rings.

Placeholder retained intentionally for investor-ready layout and future diligence evidence.

3. Competitor Analysis

The project should not compete only on headline unit price. Its defensible angle is certified quality, dependable stock, documented testing, delivery scheduling, product breadth and contractor service.



COMPETITOR LAYERS

5

informal, regional, national, cement-linked and site-cast



PRIMARY THREAT

price

informal yards can undercut if customers ignore quality



PRIMARY EDGE

reliability

consistent product, logistics and QC records

Competition is best understood as a layered market: informal local block yards, regional concrete product suppliers, national automated producers, cement-linked building-material brands and site-cast alternatives.

1

Investor lens

What the chapter means for return, risk and bankability.

2

Operating lens

Factory routines, people, quality and throughput implications.

3

Decision lens

Actions required before financial close and procurement.

Competitor Landscape

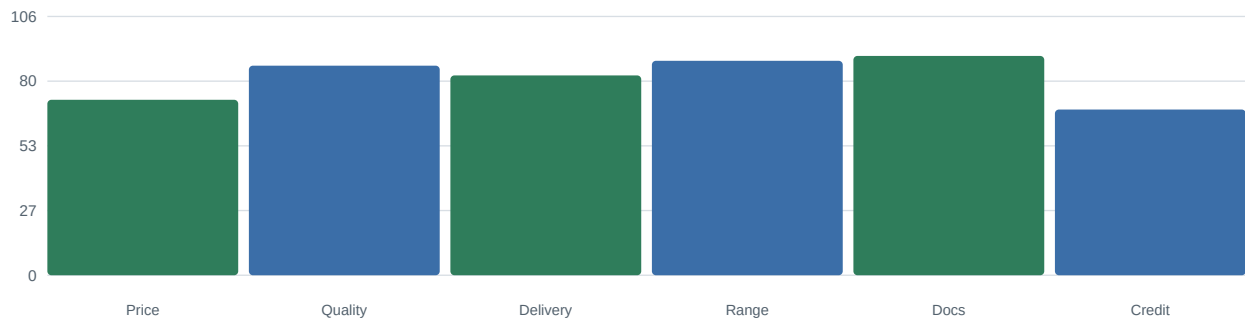
Public web references identify western Uganda suppliers such as Nicomah and national players such as Master Industries, plus cement producers with product breadth such as Hima and Tororo. This landscape confirms that buyers already understand concrete products, but quality and delivery consistency remain opportunities for differentiation.

Layer	Examples	Strength	Weakness to exploit
Informal local yards	manual block makers and small paver yards	low visible price and proximity	variable curing, dimensions and documentation
Western suppliers	Nicomah-type material suppliers	local routes and customer familiarity	may lack integrated plant depth
National automated producers	Master Industries-type suppliers	scale, product breadth and brand	freight disadvantage into western districts
Cement-linked brands	Hima/Tororo product ecosystems	cement credibility and technical know-how	not always local stock across all products
Site casting	contractor-made kerbs, channels and slabs	custom fit and perceived control	quality inconsistency and supervision burden

Competitive Positioning Map

The recommended market position is upper-middle: not the cheapest producer, but reliable enough for contractors and institutions to justify a quality premium and lower failure risk.

Illustrative competitor score by purchase criterion



Differentiation Levers

The factory should convert quality into commercial language: fewer breakages, fewer site delays, predictable dimensions, support for inspections and cleaner documentation for institutions.

Lever	Operational requirement	Buyer benefit
Stamped batch lot	lot number, date and mix family	traceability for disputes and quality assurance
Published curing standard	minimum curing days by product	reduced breakage and better confidence
Delivery booking window	dispatch board and fleet control	contractors plan labor with less idle time
Contractor technical sheet	product specs, use cases, installation notes	reduces misuse and callbacks
Credit scorecard	finance-approved terms	protects cash while supporting repeat customers

Competitor Response Scenarios

Competitor response should be expected after launch, especially if the factory gains institutional contracts. The response plan should be prepared before discounts start.

Scenario	Likely trigger	Response
Price undercut	large block order	hold list price; offer delivery reliability and quality documentation, not blind discounting
Claims on strength	contractor dispute	produce cube results, retained samples and batch ticket record
Dealer lock-in	route expansion	target unserved dealers and direct contractors before exclusive dealer deals
Fast imitation	popular paver profile	rotate colors, introduce installation support and maintain mold condition
Credit pressure	contractor demands terms	tie credit to signed acceptance, aging performance and project owner reputation

Competitor Due Diligence Checklist

Before final approval, the sponsor should complete a field competitor survey. The goal is to identify delivered prices, payment terms, breakage rates, stockouts and product profiles actually available in western Uganda.

Check	Question	Evidence
Price	What is actual delivered price by zone?	quote screenshots, receipts and contractor interviews
Quality	What products break or vary most?	site photos and mason feedback
Lead time	How often do buyers wait?	contractor procurement logs
Credit	Who offers terms and to whom?	dealer interviews and payment practice
Range	Which profiles are hard to source?	dealer shelves and project BOQs
Standards	Who can show test data?	product certificates and lab records

Competitor Profile: Informal block makers

Informal block makers should be tracked as a separate competitor group because buyer behavior, price flexibility and service promise differ materially from the other groups.

The factory's response should be evidence-led: compare total installed cost, breakage, delivery risk and documentation, not just ex-yard unit price.

Dimension	Assessment
Customer pull	high for routine products, moderate for premium profiles
Likely price behavior	discounting around large tenders and seasonal slowdowns
Operational watchpoint	stock availability, cure time, product damage and delivery reliability
Recommended response	quality files, reliable loading, zone-based delivery and disciplined credit

Competitor Profile: Western material suppliers

Western material suppliers should be tracked as a separate competitor group because buyer behavior, price flexibility and service promise differ materially from the other groups.

The factory's response should be evidence-led: compare total installed cost, breakage, delivery risk and documentation, not just ex-yard unit price.

Dimension	Assessment
Customer pull	high for routine products, moderate for premium profiles
Likely price behavior	discounting around large tenders and seasonal slowdowns
Operational watchpoint	stock availability, cure time, product damage and delivery reliability
Recommended response	quality files, reliable loading, zone-based delivery and disciplined credit

Competitor Profile: National automated producers

National automated producers should be tracked as a separate competitor group because buyer behavior, price flexibility and service promise differ materially from the other groups.

The factory's response should be evidence-led: compare total installed cost, breakage, delivery risk and documentation, not just ex-yard unit price.

Dimension	Assessment
Customer pull	high for routine products, moderate for premium profiles
Likely price behavior	discounting around large tenders and seasonal slowdowns
Operational watchpoint	stock availability, cure time, product damage and delivery reliability
Recommended response	quality files, reliable loading, zone-based delivery and disciplined credit

Competitor Profile: Cement-linked producers

Cement-linked producers should be tracked as a separate competitor group because buyer behavior, price flexibility and service promise differ materially from the other groups.

The factory's response should be evidence-led: compare total installed cost, breakage, delivery risk and documentation, not just ex-yard unit price.

Dimension	Assessment
Customer pull	high for routine products, moderate for premium profiles
Likely price behavior	discounting around large tenders and seasonal slowdowns
Operational watchpoint	stock availability, cure time, product damage and delivery reliability
Recommended response	quality files, reliable loading, zone-based delivery and disciplined credit

4. Product Portfolio

The initial production plan should prioritize blocks and standard pavers for utilization, while maintaining molds and process capability for kerbs, channels, culverts and ready-mix as contract opportunities emerge.

PRODUCT FAMILIES

13

all requested products included

CORE VOLUME

Blocks

daily builder demand and dealer replenishment

MARGIN DEPTH

Pavers/civils

premium profiles and infrastructure applications

The product portfolio combines high-frequency masonry units with higher-margin paving and infrastructure products. This gives the factory multiple demand cycles and reduces reliance on one buyer segment.

1 Investor lens

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2 Operating lens

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3 Decision lens

Actions required before financial close and procurement.

Portfolio Master Table

Indicative prices are model inputs for screening only and should be refreshed using Mbarara delivered quotations before financing. Pavers are priced by square metre; ready-mix by cubic metre; culverts and kerbs by unit.

Product	Code	Format	Primary buyers	Indicative unit price
4-inch hollow blocks	B100	100 x 200 x 400 mm	internal partitions, low-load walls, fast estate infill	UGX 1,900
6-inch hollow blocks	B150	150 x 200 x 400 mm	external walls, boundary walls, schools, clinics	UGX 2,600
8-inch hollow blocks	B200	200 x 200 x 400 mm	retaining, commercial walls, institutional projects	UGX 3,400
solid concrete blocks	BSOL	100/150/200 mm families	load-bearing details, plinths, hard-wearing walls	UGX 3,200
zig-zag pavers	PZZ	60/80 mm thickness	driveways, parking, walkways and estates	UGX 43,000
double-T pavers	PDT	60/80 mm thickness	heavy interlocking yards, schools and stations	UGX 45,500
hexagonal pavers	PHX	60 mm standard	landscape, hotels, pedestrian precincts	UGX 47,000
cobblestone pavers	PCB	60/80 mm aesthetic setts	premium courtyards, resorts, walkways	UGX 52,000
grass/permeable pavers	PGP	80/100 mm permeable grid	green parking, stormwater-sensitive sites	UGX 56,000
kerbstones	KRB	1000 mm road profiles	roads, estates, car parks and drainage edges	UGX 23,500
drainage channels	DCH	600/1000 mm channel units	roads, industrial drainage, estates	UGX 78,000
culverts	CLV	300-1200 mm diameter	rural roads, city drainage, farm crossings	UGX 275,000
ready-mix concrete	RMC	M15-M40 supplied by m3	commercial slabs, foundations, infrastructure	UGX 365,000

Portfolio Role Matrix

The role of each product is different. A healthy factory should not optimize only for output volume; it should balance utilization, cash conversion, margin and customer acquisition.

Portfolio role	Products	Commercial purpose	Management metric
Traffic builder	4-inch and 6-inch hollow blocks	frequent sales and dealer pull	daily units sold and breakage
Specification anchor	8-inch blocks, solid blocks	institutional and structural applications	cube pass rate and dimensions
Margin enhancer	double-T, hexagonal, cobblestone pavers	premium landscape and parking work	gross margin by profile
Infrastructure ticket	kerbs, channels, culverts	road and drainage contracts	tender win rate and delivery completion
Service platform	ready-mix concrete	commercial pours and large sites	on-time pours and rejected loads

Product Sheet: 4-inch hollow blocks

4-inch hollow blocks (B100) is modeled as a formal factory product with standard format of 100 x 200 x 400 mm. The primary market is internal partitions, low-load walls, fast estate infill.

Indicative strength basis: 3.5-5.0 MPa target subject to UNBS testing. Indicative mix starting point: 1 cement : 6-8 graded sand/stone dust. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

Attribute	Investor-grade operating note
Target customers	internal partitions, low-load walls, fast estate infill
Indicative unit price	UGX 1,900
Indicative daily capacity	5,200 units or m2/m3 basis as applicable
Quality control	dimension checks, batch records, visual defects, curing status and compressive strength testing
Sales note	quote ex-works and delivered-zone prices separately; enforce payment terms by customer class

PRODUCT IMAGE PLACEHOLDER

Insert product photo/rendering for 4-inch hollow blocks.

Product Sheet: 6-inch hollow blocks

6-inch hollow blocks (B150) is modeled as a formal factory product with standard format of 150 x 200 x 400 mm. The primary market is external walls, boundary walls, schools, clinics.

Indicative strength basis: 5.0-7.5 MPa target. Indicative mix starting point: 1 cement : 5.5-7 aggregate blend. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

Attribute	Investor-grade operating note
Target customers	external walls, boundary walls, schools, clinics
Indicative unit price	UGX 2,600
Indicative daily capacity	4,300 units or m2/m3 basis as applicable
Quality control	dimension checks, batch records, visual defects, curing status and compressive strength testing
Sales note	quote ex-works and delivered-zone prices separately; enforce payment terms by customer class

PRODUCT IMAGE PLACEHOLDER

Insert product photo/rendering for 6-inch hollow blocks.

Product Sheet: 8-inch hollow blocks

8-inch hollow blocks (B200) is modeled as a formal factory product with standard format of 200 x 200 x 400 mm. The primary market is retaining, commercial walls, institutional projects.

Indicative strength basis: 7.5-10.0 MPa target. Indicative mix starting point: 1 cement : 5-6.5 aggregate blend. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

Attribute	Investor-grade operating note
Target customers	retaining, commercial walls, institutional projects
Indicative unit price	UGX 3,400
Indicative daily capacity	3,300 units or m2/m3 basis as applicable
Quality control	dimension checks, batch records, visual defects, curing status and compressive strength testing
Sales note	quote ex-works and delivered-zone prices separately; enforce payment terms by customer class

PRODUCT IMAGE PLACEHOLDER

Insert product photo/rendering for 8-inch hollow blocks.

Product Sheet: solid concrete blocks

solid concrete blocks (BSOL) is modeled as a formal factory product with standard format of 100/150/200 mm families. The primary market is load-bearing details, plinths, hard-wearing walls.

Indicative strength basis: 7.5-12.5 MPa target. Indicative mix starting point: 1 cement : 4.5-6 aggregate blend. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

Attribute	Investor-grade operating note
Target customers	load-bearing details, plinths, hard-wearing walls
Indicative unit price	UGX 3,200
Indicative daily capacity	2,600 units or m2/m3 basis as applicable
Quality control	dimension checks, batch records, visual defects, curing status and compressive strength testing
Sales note	quote ex-works and delivered-zone prices separately; enforce payment terms by customer class

PRODUCT IMAGE PLACEHOLDER

Insert product photo/rendering for solid concrete blocks.

Product Sheet: zig-zag pavers

zig-zag pavers (PZZ) is modeled as a formal factory product with standard format of 60/80 mm thickness. The primary market is driveways, parking, walkways and estates.

Indicative strength basis: 35-45 MPa target. Indicative mix starting point: 1 cement : 2 sand : 3 aggregate. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

Attribute	Investor-grade operating note
Target customers	driveways, parking, walkways and estates
Indicative unit price	UGX 43,000
Indicative daily capacity	780 units or m2/m3 basis as applicable
Quality control	dimension checks, batch records, visual defects, curing status and compressive strength testing
Sales note	quote ex-works and delivered-zone prices separately; enforce payment terms by customer class

PRODUCT IMAGE PLACEHOLDER

Insert product photo/rendering for zig-zag pavers.

Product Sheet: double-T pavers

double-T pavers (PDT) is modeled as a formal factory product with standard format of 60/80 mm thickness. The primary market is heavy interlocking yards, schools and stations.

Indicative strength basis: 40-50 MPa target. Indicative mix starting point: 1 cement : 1.8 sand : 2.7 aggregate. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

Attribute	Investor-grade operating note
Target customers	heavy interlocking yards, schools and stations
Indicative unit price	UGX 45,500
Indicative daily capacity	690 units or m2/m3 basis as applicable
Quality control	dimension checks, batch records, visual defects, curing status and compressive strength testing
Sales note	quote ex-works and delivered-zone prices separately; enforce payment terms by customer class

PRODUCT IMAGE PLACEHOLDER

Insert product photo/rendering for double-T pavers.

Product Sheet: hexagonal pavers

hexagonal pavers (PHX) is modeled as a formal factory product with standard format of 60 mm standard. The primary market is landscape, hotels, pedestrian precincts.

Indicative strength basis: 35-45 MPa target. Indicative mix starting point: 1 cement : 2 sand : 3 aggregate. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

Attribute	Investor-grade operating note
Target customers	landscape, hotels, pedestrian precincts
Indicative unit price	UGX 47,000
Indicative daily capacity	510 units or m2/m3 basis as applicable
Quality control	dimension checks, batch records, visual defects, curing status and compressive strength testing
Sales note	quote ex-works and delivered-zone prices separately; enforce payment terms by customer class

PRODUCT IMAGE PLACEHOLDER

Insert product photo/rendering for hexagonal pavers.

Product Sheet: cobblestone pavers

cobblestone pavers (PCB) is modeled as a formal factory product with standard format of 60/80 mm aesthetic setts. The primary market is premium courtyards, resorts, walkways.

Indicative strength basis: 35-45 MPa target. Indicative mix starting point: 1 cement : 1.8 sand : 2.8 aggregate. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

Attribute	Investor-grade operating note
Target customers	premium courtyards, resorts, walkways
Indicative unit price	UGX 52,000
Indicative daily capacity	390 units or m2/m3 basis as applicable
Quality control	dimension checks, batch records, visual defects, curing status and compressive strength testing
Sales note	quote ex-works and delivered-zone prices separately; enforce payment terms by customer class

PRODUCT IMAGE PLACEHOLDER

Insert product photo/rendering for cobblestone pavers.

Product Sheet: grass/permeable pavers

grass/permeable pavers (PGP) is modeled as a formal factory product with standard format of 80/100 mm permeable grid. The primary market is green parking, stormwater-sensitive sites.

Indicative strength basis: 30-40 MPa target with void control. Indicative mix starting point: gap-graded aggregate with controlled paste. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

Attribute	Investor-grade operating note
Target customers	green parking, stormwater-sensitive sites
Indicative unit price	UGX 56,000
Indicative daily capacity	300 units or m2/m3 basis as applicable
Quality control	dimension checks, batch records, visual defects, curing status and compressive strength testing
Sales note	quote ex-works and delivered-zone prices separately; enforce payment terms by customer class

PRODUCT IMAGE PLACEHOLDER

Insert product photo/rendering for grass/permeable pavers.

Product Sheet: kerbstones

kerbstones (KRB) is modeled as a formal factory product with standard format of 1000 mm road profiles. The primary market is roads, estates, car parks and drainage edges.

Indicative strength basis: M25-M35 concrete target. Indicative mix starting point: M25-M35 design mix with reinforcement where required. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

Attribute	Investor-grade operating note
Target customers	roads, estates, car parks and drainage edges
Indicative unit price	UGX 23,500
Indicative daily capacity	560 units or m2/m3 basis as applicable
Quality control	dimension checks, batch records, visual defects, curing status and compressive strength testing
Sales note	quote ex-works and delivered-zone prices separately; enforce payment terms by customer class

PRODUCT IMAGE PLACEHOLDER

Insert product photo/rendering for kerbstones.

Product Sheet: drainage channels

drainage channels (DCH) is modeled as a formal factory product with standard format of 600/1000 mm channel units. The primary market is roads, industrial drainage, estates.

Indicative strength basis: M25-M35 reinforced concrete. Indicative mix starting point: M25-M35 design mix; steel cages as specified. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

Attribute	Investor-grade operating note
Target customers	roads, industrial drainage, estates
Indicative unit price	UGX 78,000
Indicative daily capacity	95 units or m2/m3 basis as applicable
Quality control	dimension checks, batch records, visual defects, curing status and compressive strength testing
Sales note	quote ex-works and delivered-zone prices separately; enforce payment terms by customer class

PRODUCT IMAGE PLACEHOLDER

Insert product photo/rendering for drainage channels.

Product Sheet: culverts

culverts (CLV) is modeled as a formal factory product with standard format of 300-1200 mm diameter. The primary market is rural roads, city drainage, farm crossings.

Indicative strength basis: reinforced concrete pipe class by design. Indicative mix starting point: M30+ with cage reinforcement and steam/wet curing option. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

Attribute	Investor-grade operating note
Target customers	rural roads, city drainage, farm crossings
Indicative unit price	UGX 275,000
Indicative daily capacity	25 units or m2/m3 basis as applicable
Quality control	dimension checks, batch records, visual defects, curing status and compressive strength testing
Sales note	quote ex-works and delivered-zone prices separately; enforce payment terms by customer class

PRODUCT IMAGE PLACEHOLDER

Insert product photo/rendering for culverts.

Product Sheet: ready-mix concrete

ready-mix concrete (RMC) is modeled as a formal factory product with standard format of M15-M40 supplied by m3. The primary market is commercial slabs, foundations, infrastructure.

Indicative strength basis: designed by grade with cube verification. Indicative mix starting point: approved design mix by grade and slump. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

Attribute	Investor-grade operating note
Target customers	commercial slabs, foundations, infrastructure
Indicative unit price	UGX 365,000
Indicative daily capacity	95 units or m2/m3 basis as applicable
Quality control	dimension checks, batch records, visual defects, curing status and compressive strength testing
Sales note	quote ex-works and delivered-zone prices separately; enforce payment terms by customer class

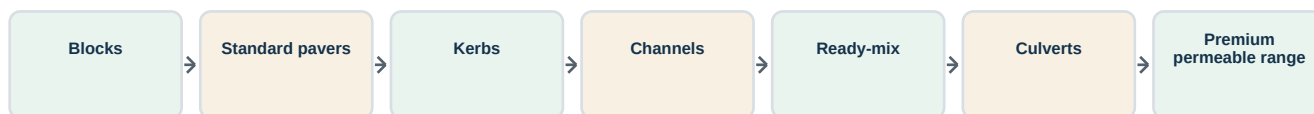
PRODUCT IMAGE PLACEHOLDER

Insert product photo/rendering for ready-mix concrete.

Portfolio Launch Sequence

The launch sequence should favor products that stabilize the core production line before expanding into highly scheduled or engineered products.

Recommended product sequencing



Stage-gate rule: do not launch the next family until batch records, curing compliance and receivable days remain within management thresholds for two consecutive months.

5. Mix Ratios

The NBRB concrete guidance referenced in this report emphasizes controlled water-cement ratio, cube sampling, curing and mechanical compaction. These principles are built into the proposed factory quality plan.

WATER-CEMENT CONTROL

≤ 0.50

NBRB guidance for dry aggregates

CUBE SAMPLING

6 per mix

3 tested at 7 days and 3 at 28 days

CURING TARGET

14 days

minimum curing reference from NBRB guidance

Mix ratios in this report are starting points for feasibility and mold trials, not final engineering specifications. Final ratios must be approved through controlled batch testing and strength results.

1 Investor lens

What the chapter means for return, risk and bankability.

2 Operating lens

Factory routines, people, quality and throughput implications.

3 Decision lens

Actions required before financial close and procurement.

Mix Ratio Summary

The factory should move from volume ratios to weigh-batched recipes once production stabilizes. Moisture correction is critical because sand and stone dust moisture directly affects strength, texture and demolding.

Product	Indicative mix by volume	Water control	QC trigger
4-inch hollow blocks	1 cement : 6-8 graded sand/stone dust	target w/c <= 0.50 where applicable	trial cubes and dimensional pass
6-inch hollow blocks	1 cement : 5.5-7 aggregate blend	target w/c <= 0.50 where applicable	trial cubes and dimensional pass
8-inch hollow blocks	1 cement : 5-6.5 aggregate blend	target w/c <= 0.50 where applicable	trial cubes and dimensional pass
solid concrete blocks	1 cement : 4.5-6 aggregate blend	target w/c <= 0.50 where applicable	trial cubes and dimensional pass
zig-zag pavers	1 cement : 2 sand : 3 aggregate	target w/c <= 0.50 where applicable	trial cubes and dimensional pass
double-T pavers	1 cement : 1.8 sand : 2.7 aggregate	target w/c <= 0.50 where applicable	trial cubes and dimensional pass
hexagonal pavers	1 cement : 2 sand : 3 aggregate	target w/c <= 0.50 where applicable	trial cubes and dimensional pass
cobblestone pavers	1 cement : 1.8 sand : 2.8 aggregate	target w/c <= 0.50 where applicable	trial cubes and dimensional pass
grass/permeable pavers	gap-graded aggregate with controlled paste	target w/c <= 0.50 where applicable	trial cubes and dimensional pass
kerbstones	M25-M35 design mix with reinforcement where required	target w/c <= 0.50 where applicable	trial cubes and dimensional pass
drainage channels	M25-M35 design mix; steel cages as specified	target w/c <= 0.50 where applicable	trial cubes and dimensional pass
culverts	M30+ with cage reinforcement and steam/wet curing option	target w/c <= 0.50 where applicable	trial cubes and dimensional pass
ready-mix concrete	approved design mix by grade and slump	target w/c <= 0.50 where applicable	trial cubes and dimensional pass

Mix Design Trial Sheet: Masonry Blocks

Products covered: 4-inch hollow blocks, 6-inch hollow blocks, 8-inch hollow blocks and solid concrete blocks. Starting mix basis: 1 cement : 4.5-8 aggregate blend depending on strength target and unit thickness. This should be trialed across at least three moisture conditions and two cement sources before production sign-off.

The acceptance gate should combine machine behavior, surface finish, demolding quality, curing result, dimensional tolerance and compressive strength. A ratio that looks cheaper but increases breakage is not acceptable.

Trial field	Entry
Cement type	CEM class to be recorded by supplier batch
Aggregate blend	sand, stone dust and coarse aggregate proportions by weight
Water addition	metered water plus moisture adjustment
Admixture	only after baseline mix is proven
Cube/testing plan	minimum 7-day and 28-day checks; retain batch samples
QC focus	prioritize dimensional tolerance, demolding quality, breakage rate and 7/28-day cube trend
Production sign-off	plant manager, QC lead and maintenance lead

Mix Design Trial Sheet: Standard Pavers

Products covered: zig-zag, double-T and hexagonal pavers. Starting mix basis: 1 cement : 1.8-2 sand : 2.7-3 aggregate with controlled water and pigment only after baseline approval. This should be trialed across at least three moisture conditions and two cement sources before production sign-off.

The acceptance gate should combine machine behavior, surface finish, demolding quality, curing result, dimensional tolerance and compressive strength. A ratio that looks cheaper but increases breakage is not acceptable.

Trial field	Entry
Cement type	CEM class to be recorded by supplier batch
Aggregate blend	sand, stone dust and coarse aggregate proportions by weight
Water addition	metered water plus moisture adjustment
Admixture	only after baseline mix is proven
Cube/testing plan	minimum 7-day and 28-day checks; retain batch samples
QC focus	prioritize surface finish, abrasion resistance, interlock fit and color consistency
Production sign-off	plant manager, QC lead and maintenance lead

Mix Design Trial Sheet: Premium and Permeable Pavers

Products covered: cobblestone and grass/permeable pavers. Starting mix basis: high-strength paver mix or gap-graded permeable mix based on void and drainage target. This should be trialed across at least three moisture conditions and two cement sources before production sign-off.

The acceptance gate should combine machine behavior, surface finish, demolding quality, curing result, dimensional tolerance and compressive strength. A ratio that looks cheaper but increases breakage is not acceptable.

Trial field	Entry
Cement type	CEM class to be recorded by supplier batch
Aggregate blend	sand, stone dust and coarse aggregate proportions by weight
Water addition	metered water plus moisture adjustment
Admixture	only after baseline mix is proven
Cube/testing plan	minimum 7-day and 28-day checks; retain batch samples
QC focus	prioritize void control, edge durability, installed appearance and customer installation guidance
Production sign-off	plant manager, QC lead and maintenance lead

Mix Design Trial Sheet: Civil Precast Products

Products covered: kerbstones, drainage channels and culverts. Starting mix basis: M25-M35+ design mixes with reinforcement where specified. This should be trialed across at least three moisture conditions and two cement sources before production sign-off.

The acceptance gate should combine machine behavior, surface finish, demolding quality, curing result, dimensional tolerance and compressive strength. A ratio that looks cheaper but increases breakage is not acceptable.

Trial field	Entry
Cement type	CEM class to be recorded by supplier batch
Aggregate blend	sand, stone dust and coarse aggregate proportions by weight
Water addition	metered water plus moisture adjustment
Admixture	only after baseline mix is proven
Cube/testing plan	minimum 7-day and 28-day checks; retain batch samples
QC focus	prioritize cage placement, cover, compaction, curing and dimensional checks
Production sign-off	plant manager, QC lead and maintenance lead

Mix Design Trial Sheet: Ready-Mix Concrete

Products covered: M15-M40 supplied by cubic metre. Starting mix basis: engineer-approved design mix by grade, slump and aggregate size. This should be trialed across at least three moisture conditions and two cement sources before production sign-off.

The acceptance gate should combine machine behavior, surface finish, demolding quality, curing result, dimensional tolerance and compressive strength. A ratio that looks cheaper but increases breakage is not acceptable.

Trial field	Entry
Cement type	CEM class to be recorded by supplier batch
Aggregate blend	sand, stone dust and coarse aggregate proportions by weight
Water addition	metered water plus moisture adjustment
Admixture	only after baseline mix is proven
Cube/testing plan	minimum 7-day and 28-day checks; retain batch samples
QC focus	prioritize batch tickets, slump, temperature where relevant, site sign-off and cube records
Production sign-off	plant manager, QC lead and maintenance lead

Ready-Mix Concrete Design Families

Ready-mix should be sold by grade, slump and application. Each design family should have a locked recipe, permitted adjustment range, batch ticket and site acceptance protocol.

Grade	Typical applications	Design focus	QC checks
M15	blinding, non-structural works	workability and cost control	slump, batch ticket
M20	light slabs and foundations	strength consistency	slump and cubes
M25	commercial slabs and beams	cement efficiency and placement	slump, cubes, temperature
M30+	specified structures	engineering approval	formal mix approval and cube program

All RMC mixes must be confirmed by a qualified engineer and site-specific specifications before sale.

6. Raw Materials

The procurement strategy should use dual sourcing where practical, moisture-controlled storage, supplier scorecards and batch traceability from receipt to dispatch.

CRITICAL INPUT

Cement

price, quality and availability dominate gross margin

HIDDEN VARIABLE

Moisture

sand and dust moisture can undermine recipes

STOCK POLICY

14-30 days

depends on product and cement price volatility

Raw material control is the biggest operational determinant of quality and margin. Cement, aggregates, sand, water, steel, pigments and admixtures must be managed as production inputs, not just purchases.

1 Investor lens

What the chapter means for return, risk and bankability.

2 Operating lens

Factory routines, people, quality and throughput implications.

3 Decision lens

Actions required before financial close and procurement.

Raw Material Control Register

Every raw material should have a purchase specification, receipt inspection, storage rule and rejection procedure. A cheap material that creates rejects is a margin leak.

Input	Specification	Storage/control	Risk
Cement	CEM I/II or grade required by product	silo and bag storage; FIFO; certificates	price volatility and supply interruption
Washed sand	clean, graded, low clay/silt	covered bays and moisture checks	strength loss and surface defects
Stone dust	consistent grading	separate from sand; avoid contamination	water demand and dimensional variation
Coarse aggregate	6-20 mm as mix requires	segregated bins; loader discipline	honeycombing and poor compaction
Water	potable or approved production water	tank, meter, recycling controls	strength and efflorescence issues
Steel reinforcement	certified bars/wire cages	dry storage and certificates	culvert/channel structural risk
Pigments/admixtures	approved supplier and dosage	locked dosing and trial records	color mismatch and setting variability
Pallets/molds release	machine-compatible consumables	maintenance register	surface defects and downtime

Material Sheet: Cement

Specification focus: CEM I/II or grade required by product. Control requirement: silo and bag storage; FIFO; certificates. Key risk: price volatility and supply interruption.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

Control point	Procedure
Supplier qualification	collect certificates, route reliability and delivered-cost history
Receipt inspection	visual check, weight/volume reconciliation and contamination check
Storage	silo and bag storage; FIFO; certificates
Production release	QC or supervisor approval before use
Monthly scorecard	price, delivery reliability, rejection rate and complaint linkage

Material Sheet: Washed sand

Specification focus: clean, graded, low clay/silt. Control requirement: covered bays and moisture checks. Key risk: strength loss and surface defects.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

Control point	Procedure
Supplier qualification	collect certificates, route reliability and delivered-cost history
Receipt inspection	visual check, weight/volume reconciliation and contamination check
Storage	covered bays and moisture checks
Production release	QC or supervisor approval before use
Monthly scorecard	price, delivery reliability, rejection rate and complaint linkage

Material Sheet: Stone dust

Specification focus: consistent grading. Control requirement: separate from sand; avoid contamination. Key risk: water demand and dimensional variation.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

Control point	Procedure
Supplier qualification	collect certificates, route reliability and delivered-cost history
Receipt inspection	visual check, weight/volume reconciliation and contamination check
Storage	separate from sand; avoid contamination
Production release	QC or supervisor approval before use
Monthly scorecard	price, delivery reliability, rejection rate and complaint linkage

Material Sheet: Coarse aggregate

Specification focus: 6-20 mm as mix requires. Control requirement: segregated bins; loader discipline. Key risk: honeycombing and poor compaction.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

Control point	Procedure
Supplier qualification	collect certificates, route reliability and delivered-cost history
Receipt inspection	visual check, weight/volume reconciliation and contamination check
Storage	segregated bins; loader discipline
Production release	QC or supervisor approval before use
Monthly scorecard	price, delivery reliability, rejection rate and complaint linkage

Material Sheet: Water

Specification focus: potable or approved production water. Control requirement: tank, meter, recycling controls. Key risk: strength and efflorescence issues.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

Control point	Procedure
Supplier qualification	collect certificates, route reliability and delivered-cost history
Receipt inspection	visual check, weight/volume reconciliation and contamination check
Storage	tank, meter, recycling controls
Production release	QC or supervisor approval before use
Monthly scorecard	price, delivery reliability, rejection rate and complaint linkage

Material Sheet: Steel reinforcement

Specification focus: certified bars/wire cages. Control requirement: dry storage and certificates. Key risk: culvert/channel structural risk.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

Control point	Procedure
Supplier qualification	collect certificates, route reliability and delivered-cost history
Receipt inspection	visual check, weight/volume reconciliation and contamination check
Storage	dry storage and certificates
Production release	QC or supervisor approval before use
Monthly scorecard	price, delivery reliability, rejection rate and complaint linkage

Material Sheet: Pigments/admixtures

Specification focus: approved supplier and dosage. Control requirement: locked dosing and trial records. Key risk: color mismatch and setting variability.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

Control point	Procedure
Supplier qualification	collect certificates, route reliability and delivered-cost history
Receipt inspection	visual check, weight/volume reconciliation and contamination check
Storage	locked dosing and trial records
Production release	QC or supervisor approval before use
Monthly scorecard	price, delivery reliability, rejection rate and complaint linkage

Material Sheet: Pallets/molds release

Specification focus: machine-compatible consumables. Control requirement: maintenance register. Key risk: surface defects and downtime.

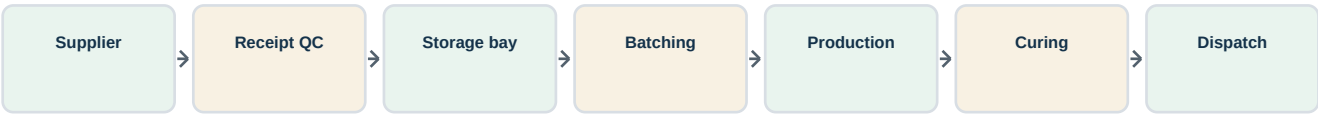
Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

Control point	Procedure
Supplier qualification	collect certificates, route reliability and delivered-cost history
Receipt inspection	visual check, weight/volume reconciliation and contamination check
Storage	maintenance register
Production release	QC or supervisor approval before use
Monthly scorecard	price, delivery reliability, rejection rate and complaint linkage

Supply Chain Diagram

The factory should minimize double handling. Aggregates and sand flow to covered bays, cement to silos or dry stores, water through metering and recycling, and finished products through curing to yard zones.

Raw material to finished goods control flow



7. Machinery Options

The investment should avoid under-buying the core line and then losing margin through poor consistency, high rejects and downtime. However, ready-mix and culvert capacity should be phased by customer evidence.

RECOMMENDED

Option B

automatic vibropress core line

KEY PROCUREMENT RISK

Support

local service, spares and commissioning discipline

MOLD STRATEGY

modular

blocks, pavers, kerbs and premium profiles

Machinery selection determines quality, labor intensity, mold flexibility, downtime risk and expansion potential. The recommended base case is a modular automatic line with local service planning and a realistic spares pack.

1 Investor lens

What the chapter means for return, risk and bankability.

2 Operating lens

Factory routines, people, quality and throughput implications.

3 Decision lens

Actions required before financial close and procurement.

Machinery Option Comparison

Option B is the recommended base-case line because it offers credible quality and capacity without overcommitting to a national-scale plant before demand is proven.

Option	Description	Indicative CAPEX	Pros	Cons
A	Semi-automatic block/paver line	UGX 5.80 bn	lower entry cost; easier staffing	higher labor, lower consistency, slower scale
B	Automatic vibropress with batching and stacker	UGX 9.40 bn	recommended quality and throughput	higher funding and maintenance discipline
C	High-capacity integrated plant with robotics	UGX 15.50 bn	best scale economics	too much demand risk before anchor contracts
D	Used/refurbished imported line	UGX 4.90 bn	lower headline cost	hidden downtime, spares and controls risk

Core Vibropress Line

Scope: automatic block/paver machine, board feed, stacker and mold set. Investor value: dimension consistency, fast mold changeover and vibration control.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

Procurement requirement	Reason
Factory acceptance test	confirms machine function before shipment
Site acceptance test	confirms installed throughput and product quality
Critical spares pack	protects first-year uptime
Operator training	reduces rejects and maintenance abuse
Mold documentation	ensures dimensions, wear limits and replacement planning

Batching and Mixing

Scope: aggregate bins, cement weighing, pan mixer and water meter. Investor value: recipe repeatability and moisture correction.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

Procurement requirement	Reason
Factory acceptance test	confirms machine function before shipment
Site acceptance test	confirms installed throughput and product quality
Critical spares pack	protects first-year uptime
Operator training	reduces rejects and maintenance abuse
Mold documentation	ensures dimensions, wear limits and replacement planning

Curing System

Scope: covered racks, water misting, chambers and product lot control. Investor value: strength development and reduced breakage.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

Procurement requirement	Reason
Factory acceptance test	confirms machine function before shipment
Site acceptance test	confirms installed throughput and product quality
Critical spares pack	protects first-year uptime
Operator training	reduces rejects and maintenance abuse
Mold documentation	ensures dimensions, wear limits and replacement planning

RMC Plant

Scope: 30-60 m³/h batching plant, silos, controls and transit mixers. Investor value: commercial concrete supply and site pours.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

Procurement requirement	Reason
Factory acceptance test	confirms machine function before shipment
Site acceptance test	confirms installed throughput and product quality
Critical spares pack	protects first-year uptime
Operator training	reduces rejects and maintenance abuse
Mold documentation	ensures dimensions, wear limits and replacement planning

Culvert and Drainage Equipment

Scope: pipe molds, cage rollers, vibrators, cranes and curing area. Investor value: road/drainage tender capability.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

Procurement requirement	Reason
Factory acceptance test	confirms machine function before shipment
Site acceptance test	confirms installed throughput and product quality
Critical spares pack	protects first-year uptime
Operator training	reduces rejects and maintenance abuse
Mold documentation	ensures dimensions, wear limits and replacement planning

Material Handling

Scope: wheel loader, forklifts, pallet trucks, tippers and flatbeds. Investor value: yard productivity and delivery reliability.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

Procurement requirement	Reason
Factory acceptance test	confirms machine function before shipment
Site acceptance test	confirms installed throughput and product quality
Critical spares pack	protects first-year uptime
Operator training	reduces rejects and maintenance abuse
Mold documentation	ensures dimensions, wear limits and replacement planning

Power Backup

Scope: generator, transfer switch and critical load plan. Investor value: reduced downtime from outages.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

Procurement requirement	Reason
Factory acceptance test	confirms machine function before shipment
Site acceptance test	confirms installed throughput and product quality
Critical spares pack	protects first-year uptime
Operator training	reduces rejects and maintenance abuse
Mold documentation	ensures dimensions, wear limits and replacement planning

Weighbridge and ERP

Scope: truck weighbridge, batch tickets, inventory and invoicing. Investor value: shrinkage control and finance visibility.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

Procurement requirement	Reason
Factory acceptance test	confirms machine function before shipment
Site acceptance test	confirms installed throughput and product quality
Critical spares pack	protects first-year uptime
Operator training	reduces rejects and maintenance abuse
Mold documentation	ensures dimensions, wear limits and replacement planning

Machinery Procurement Gate

No equipment order should be released until civil foundations, power requirements, compressed air, water pressure, installation sequence and commissioning responsibility are frozen.

Machinery procurement and commissioning gate



Investor gate: retain final supplier payment until site acceptance test confirms products, throughput and controls.

8. Factory Layout

The recommended layout is a one-way material flow with segregated heavy-vehicle movement, covered curing, laboratory adjacency to production and finished goods organized by product family and delivery zone.

SITE NEED

5-8 acres

depending on stockyard depth and RMC fleet

PRIMARY FLOW

one-way

raw material to dispatch without cross-traffic

LAYOUT PRIORITY

yard control

stock, curing and dispatch discipline

The site layout must protect flow: raw materials in, controlled production, curing, quality release, finished goods storage and dispatch out. Poor layout will consume margin through double handling.

Investor lens

What the chapter means for return, risk and bankability.

Operating lens

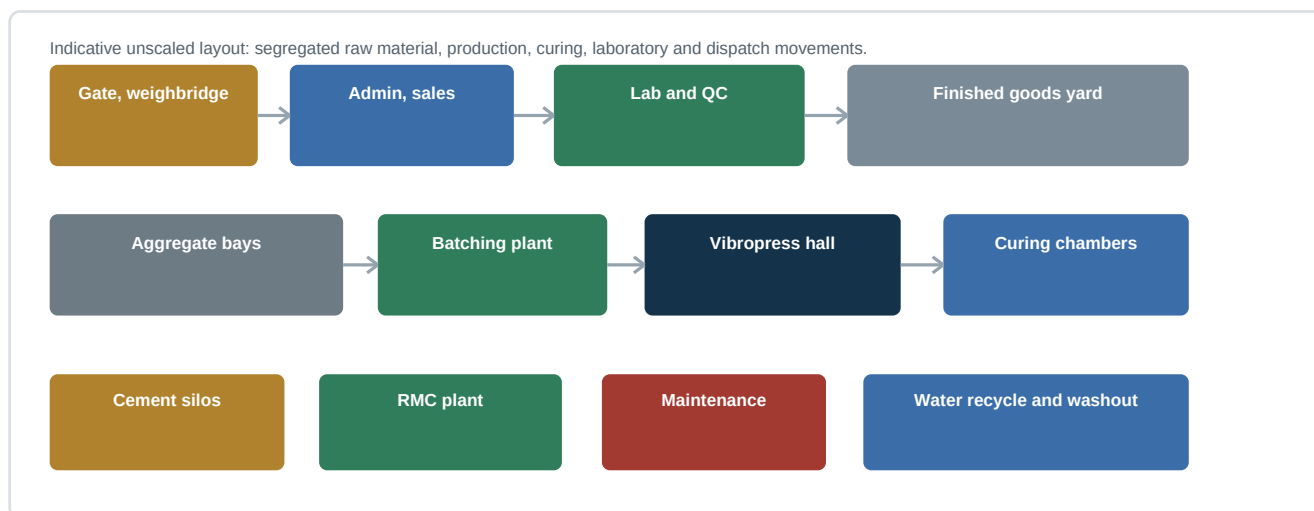
Factory routines, people, quality and throughput implications.

Decision lens

Actions required before financial close and procurement.

Indicative Factory Layout

The diagram is conceptual and unscaled. The final layout should be completed by an industrial engineer after geotechnical results, drainage design, truck turning radii and utility routes are known.



Layout Zone: Gate and Weighbridge

Control all inbound raw materials and outbound product loads; capture truck, supplier, product, weight and document number.

Design principle: vehicle movement, people movement and material movement should be visible, signed and separated where practical. The layout should make the correct operating behavior the easiest behavior.

Design check	Requirement
Safety	clear walkways, PPE zones, speed limits and reversing controls
Product integrity	no mixing of green, curing and released product
Inventory	lot labels, aging status and stock count access
Drainage	falls away from production; sediment and washout management

Layout Zone: Aggregate Bays

Use separated covered bays to avoid contamination and allow moisture testing by material type.

Design principle: vehicle movement, people movement and material movement should be visible, signed and separated where practical. The layout should make the correct operating behavior the easiest behavior.

Design check	Requirement
Safety	clear walkways, PPE zones, speed limits and reversing controls
Product integrity	no mixing of green, curing and released product
Inventory	lot labels, aging status and stock count access
Drainage	falls away from production; sediment and washout management

Layout Zone: Cement Silos and Bag Store

Protect cement from moisture; design access for bulk tanker and bag deliveries without blocking production.

Design principle: vehicle movement, people movement and material movement should be visible, signed and separated where practical. The layout should make the correct operating behavior the easiest behavior.

Design check	Requirement
Safety	clear walkways, PPE zones, speed limits and reversing controls
Product integrity	no mixing of green, curing and released product
Inventory	lot labels, aging status and stock count access
Drainage	falls away from production; sediment and washout management

Layout Zone: Production Hall

Place batching, mixer, vibropress and pallet flow in a linear sequence with maintenance access.

Design principle: vehicle movement, people movement and material movement should be visible, signed and separated where practical. The layout should make the correct operating behavior the easiest behavior.

Design check	Requirement
Safety	clear walkways, PPE zones, speed limits and reversing controls
Product integrity	no mixing of green, curing and released product
Inventory	lot labels, aging status and stock count access
Drainage	falls away from production; sediment and washout management

Layout Zone: Curing Zone

Separate green products from released stock; maintain curing logs and water/mist controls.

Design principle: vehicle movement, people movement and material movement should be visible, signed and separated where practical. The layout should make the correct operating behavior the easiest behavior.

Design check	Requirement
Safety	clear walkways, PPE zones, speed limits and reversing controls
Product integrity	no mixing of green, curing and released product
Inventory	lot labels, aging status and stock count access
Drainage	falls away from production; sediment and washout management

Layout Zone: Finished Goods Yard

Organize by product, age, lot, customer reservation and dispatch route.

Design principle: vehicle movement, people movement and material movement should be visible, signed and separated where practical. The layout should make the correct operating behavior the easiest behavior.

Design check	Requirement
Safety	clear walkways, PPE zones, speed limits and reversing controls
Product integrity	no mixing of green, curing and released product
Inventory	lot labels, aging status and stock count access
Drainage	falls away from production; sediment and washout management

Layout Zone: Ready-Mix Bay

Keep RMC traffic from block dispatch queues; include washout and water recycling controls.

Design principle: vehicle movement, people movement and material movement should be visible, signed and separated where practical. The layout should make the correct operating behavior the easiest behavior.

Design check	Requirement
Safety	clear walkways, PPE zones, speed limits and reversing controls
Product integrity	no mixing of green, curing and released product
Inventory	lot labels, aging status and stock count access
Drainage	falls away from production; sediment and washout management

9. Electricity and Water Consumption

Uganda's distribution landscape changed when UEDCL assumed electricity distribution operations from Umeme in April 2025. The project should confirm connection capacity, transformer needs and reliability directly with the current distribution utility and ERA guidance.

POWER DEMAND

450-750 kVA

indicative installed load range by phase

WATER DEMAND

45-125 m3/day

depends on RMC scale, curing and recycling

CRITICAL CONTROL

metering

water, cement and power by batch/shift

Electricity and water are production risks, not back-office utilities. The plant requires reliable industrial power, backup for critical operations, metered water, recycling and batch-level water control.

1 Investor lens

What the chapter means for return, risk and bankability.

2 Operating lens

Factory routines, people, quality and throughput implications.

3 Decision lens

Actions required before financial close and procurement.

Utility Demand Model

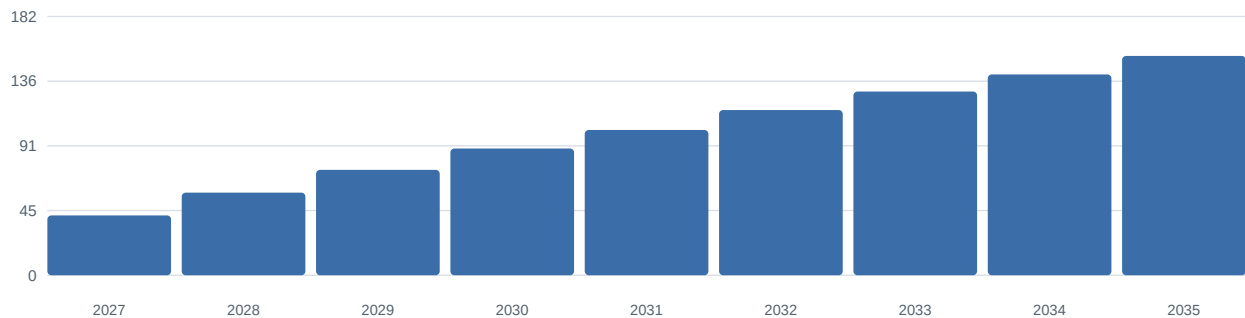
The estimates below are planning figures. Electrical engineering must confirm installed load, diversity factor, starting currents, transformer size, backup generator rating and power factor correction.

Area	Installed load	Typical operating use	Control action
Vibropress and batching	180-280 kW	80-160 kWh/hour in production	shift energy dashboard
RMC plant and pumps	80-150 kW	35-90 kWh/hour during pours	separate RMC meter
Curing, water pumps and compressors	40-90 kW	variable by curing method	pressure and water meter
Lighting/admin/security	25-45 kW	low but continuous	LED and submeter
Workshop and auxiliaries	50-120 kW	maintenance peaks	planned maintenance windows

Electricity Consumption Ramp

The base model assumes electricity intensity improves as utilization rises. The absolute bill increases with volume, but kWh per saleable unit should fall as stoppages and rejects reduce.

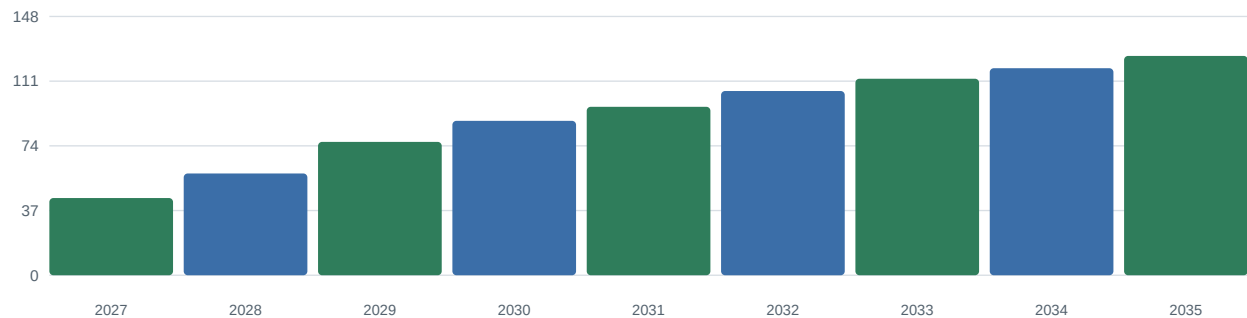
Indicative monthly electricity use by maturity stage, MWh



Water Consumption Ramp

Water use comes from mixing, curing, dust control, washout, sanitation and ready-mix production. Recycling is especially important after RMC launch because washout water and sediment can become environmental and cost risks.

Indicative average production water use, m3/day



UEDCL Connection Plan

Submit load schedule, transformer requirement, single-line diagram and site plan; confirm tariff category and service reliability expectations.

Utilities should be reviewed in the weekly management meeting because they affect cost, uptime, compliance and product quality.

KPI	Target direction
kWh per saleable unit	down as utilization and maintenance improve
Generator hours	low, exception-based
Water recycled	up with RMC and curing maturity
Unplanned utility stoppages	down with preventive maintenance

Backup Power Strategy

Back up controls, batching completion, lighting, lab and critical pumps; avoid running the entire plant on generator unless economics justify it.

Utilities should be reviewed in the weekly management meeting because they affect cost, uptime, compliance and product quality.

KPI	Target direction
kWh per saleable unit	down as utilization and maintenance improve
Generator hours	low, exception-based
Water recycled	up with RMC and curing maturity
Unplanned utility stoppages	down with preventive maintenance

Water Source Strategy

Use NWSC where available, evaluate borehole/rainwater only with permits and water quality testing, and install storage tanks for operating buffer.

Utilities should be reviewed in the weekly management meeting because they affect cost, uptime, compliance and product quality.

KPI	Target direction
kWh per saleable unit	down as utilization and maintenance improve
Generator hours	low, exception-based
Water recycled	up with RMC and curing maturity
Unplanned utility stoppages	down with preventive maintenance

Water Recycling and Washout

Separate washout, sediment traps and clarified reuse water; protect drains and maintain a visible environmental housekeeping routine.

Utilities should be reviewed in the weekly management meeting because they affect cost, uptime, compliance and product quality.

KPI	Target direction
kWh per saleable unit	down as utilization and maintenance improve
Generator hours	low, exception-based
Water recycled	up with RMC and curing maturity
Unplanned utility stoppages	down with preventive maintenance

Utility KPIs

Track kWh per unit, m3 water per batch, downtime by cause, generator hours, power factor and water recycling percentage.

Utilities should be reviewed in the weekly management meeting because they affect cost, uptime, compliance and product quality.

KPI	Target direction
kWh per saleable unit	down as utilization and maintenance improve
Generator hours	low, exception-based
Water recycled	up with RMC and curing maturity
Unplanned utility stoppages	down with preventive maintenance

10. Laboratory and Quality Control

The QC system should cover raw materials, mix recipes, machine settings, curing, dimensions, compressive strength, visual defects, release status and complaint investigation.



CUBE PROGRAM

7 and 28 days

strength development tracking



RELEASE GATE

QC sign-off

no green stock dispatched



TRACEABILITY

lot number

batch ticket to customer delivery

The laboratory is the project's trust engine. It turns concrete products from commodity blocks into documented, traceable and certifiable building materials.

1

Investor lens

What the chapter means for return, risk and bankability.

2

Operating lens

Factory routines, people, quality and throughput implications.

3

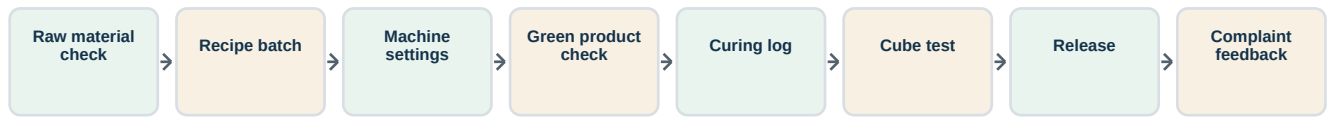
Decision lens

Actions required before financial close and procurement.

Quality Control System

NBRB guidance identifies concrete cube testing, curing and mechanical compaction as quality-control fundamentals. The factory should formalize these into daily routines with records that can be shown to customers and inspectors.

Factory QC control loop



Laboratory Equipment Register

The laboratory should be modest but credible at launch. A clean, calibrated and used lab is more valuable than an expensive lab that production ignores.

Equipment	Purpose	Control
Compression testing machine	cube strength verification	calibration certificate and safety guard
Cube molds and curing tank	7-day and 28-day testing	mold cleaning, labeling and curing water control
Sieve set and shaker	aggregate grading	weekly grading logs by supplier
Moisture meter	sand and aggregate moisture correction	shift-level moisture entries
Slump cone and rod	RMC workability	site acceptance and batch ticket match
Digital scales	weigh-batch control and lab trials	calibration and check weights
Calipers/tape/gauges	dimensional tolerance	product release checks
Sample retain area	dispute resolution	retained samples by lot and age

Incoming Material Inspection

Inspect cement certificates, sand cleanliness, aggregate grading, moisture, contamination and delivery weights before release to production.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

Record	Minimum fields
Owner	QC lead or delegated technician
Frequency	defined by product risk and production volume
Escalation	plant manager if result is out of tolerance
Retention	digital and hard-copy files by month and product

Batch Ticket Discipline

Each production batch should record recipe, operator, machine, material lots, water adjustment, start time and product family.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

Record	Minimum fields
Owner	QC lead or delegated technician
Frequency	defined by product risk and production volume
Escalation	plant manager if result is out of tolerance
Retention	digital and hard-copy files by month and product

Dimensional Tolerance Checks

Measure units at defined intervals. Reject trends usually point to mold wear, poor vibration, wrong moisture or handling damage.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

Record	Minimum fields
Owner	QC lead or delegated technician
Frequency	defined by product risk and production volume
Escalation	plant manager if result is out of tolerance
Retention	digital and hard-copy files by month and product

Curing Compliance

Log curing start, product age, water/mist conditions and release date. Green products must be physically segregated.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

Record	Minimum fields
Owner	QC lead or delegated technician
Frequency	defined by product risk and production volume
Escalation	plant manager if result is out of tolerance
Retention	digital and hard-copy files by month and product

Compressive Strength Testing

Use cube tests for design validation and routine control. Trend results by mix family and cement source.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

Record	Minimum fields
Owner	QC lead or delegated technician
Frequency	defined by product risk and production volume
Escalation	plant manager if result is out of tolerance
Retention	digital and hard-copy files by month and product

RMC Site Acceptance

For ready-mix, delivery tickets, slump results, time at site, water addition rules and sign-off protect both factory and customer.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

Record	Minimum fields
Owner	QC lead or delegated technician
Frequency	defined by product risk and production volume
Escalation	plant manager if result is out of tolerance
Retention	digital and hard-copy files by month and product

Complaint Investigation

Every complaint should trace product, batch, customer, delivery, photos, test records and corrective action.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

Record	Minimum fields
Owner	QC lead or delegated technician
Frequency	defined by product risk and production volume
Escalation	plant manager if result is out of tolerance
Retention	digital and hard-copy files by month and product

UNBS Certification Preparation

Maintain product specifications, test results, process controls and management responsibility for certification visits.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

Record	Minimum fields
Owner	QC lead or delegated technician
Frequency	defined by product risk and production volume
Escalation	plant manager if result is out of tolerance
Retention	digital and hard-copy files by month and product

11. Legal and Regulatory Offices

The compliance strategy should be a named approvals register with owners, documents, status, expiry dates and renewal reminders. Regulatory diligence should begin before land is committed.

 COMPANY

URSB

registration and legal identity

 ENVIRONMENT

NEMA

project brief or ESIA route

 PRODUCT

UNBS

standards and certification pathway

The project touches company registration, investment licensing, tax, environmental assessment, building control, product standards, labor, utilities and local trade licensing.

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Actions required before financial close and procurement.

Regulatory Office Map

The approval path should be managed as a project workstream. Missing approvals can delay commissioning, expose the sponsor to penalties and weaken lender confidence.

Office	Role	Documents/actions
URSB	company registration, legal documents	incorporation, beneficial ownership, filings
URA	TIN, VAT, PAYE, income tax	tax registration, EFRIS, returns, compliance certificate
UIA	investment license and facilitation	license application, land facilitation, aftercare
Mbarara City Council	trade license, physical planning, building approvals	zoning, building plans, local permits
NEMA	environmental screening, project brief/ESIA	ELMIS submission, approval conditions, audits
UNBS	product standards and certification	precast paving blocks, cement/product compliance
NBRB / Building Control	building code and construction oversight	factory building plans and inspections
UEDCL / ERA	electricity connection and distribution context	load application, metering, tariff and supply reliability
NWSC / Water authorities	water and sewerage connection	connection documents, meters and discharge arrangements
NSSF and labor/OHS offices	employee social security and workplace safety	registration, payroll compliance, safety records

Regulatory Sheet: URSB

Primary role: company registration, legal documents. Key documents/actions: incorporation, beneficial ownership, filings.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

Register field	Entry
Authority	URSB
Approval or compliance action	company registration, legal documents
Core file	incorporation, beneficial ownership, filings
Responsible owner	project manager during setup; compliance officer after launch
Board visibility	monthly until commissioning; quarterly after stabilization

Regulatory Sheet: URA

Primary role: TIN, VAT, PAYE, income tax. Key documents/actions: tax registration, EFRIS, returns, compliance certificate.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

Register field	Entry
Authority	URA
Approval or compliance action	TIN, VAT, PAYE, income tax
Core file	tax registration, EFRIS, returns, compliance certificate
Responsible owner	project manager during setup; compliance officer after launch
Board visibility	monthly until commissioning; quarterly after stabilization

Regulatory Sheet: UIA

Primary role: investment license and facilitation. Key documents/actions: license application, land facilitation, aftercare.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

Register field	Entry
Authority	UIA
Approval or compliance action	investment license and facilitation
Core file	license application, land facilitation, aftercare
Responsible owner	project manager during setup; compliance officer after launch
Board visibility	monthly until commissioning; quarterly after stabilization

Regulatory Sheet: Mbarara City Council

Primary role: trade license, physical planning, building approvals. Key documents/actions: zoning, building plans, local permits.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

Register field	Entry
Authority	Mbarara City Council
Approval or compliance action	trade license, physical planning, building approvals
Core file	zoning, building plans, local permits
Responsible owner	project manager during setup; compliance officer after launch
Board visibility	monthly until commissioning; quarterly after stabilization

Regulatory Sheet: NEMA

Primary role: environmental screening, project brief/ESIA. Key documents/actions: ELMIS submission, approval conditions, audits.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

Register field	Entry
Authority	NEMA
Approval or compliance action	environmental screening, project brief/ESIA
Core file	ELMIS submission, approval conditions, audits
Responsible owner	project manager during setup; compliance officer after launch
Board visibility	monthly until commissioning; quarterly after stabilization

Regulatory Sheet: UNBS

Primary role: product standards and certification. Key documents/actions: precast paving blocks, cement/product compliance.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

Register field	Entry
Authority	UNBS
Approval or compliance action	product standards and certification
Core file	precast paving blocks, cement/product compliance
Responsible owner	project manager during setup; compliance officer after launch
Board visibility	monthly until commissioning; quarterly after stabilization

Regulatory Sheet: NBRB / Building Control

Primary role: building code and construction oversight. Key documents/actions: factory building plans and inspections.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

Register field	Entry
Authority	NBRB / Building Control
Approval or compliance action	building code and construction oversight
Core file	factory building plans and inspections
Responsible owner	project manager during setup; compliance officer after launch
Board visibility	monthly until commissioning; quarterly after stabilization

Regulatory Sheet: UEDCL / ERA

Primary role: electricity connection and distribution context. Key documents/actions: load application, metering, tariff and supply reliability.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

Register field	Entry
Authority	UEDCL / ERA
Approval or compliance action	electricity connection and distribution context
Core file	load application, metering, tariff and supply reliability
Responsible owner	project manager during setup; compliance officer after launch
Board visibility	monthly until commissioning; quarterly after stabilization

Regulatory Sheet: NWSC / Water authorities

Primary role: water and sewerage connection. Key documents/actions: connection documents, meters and discharge arrangements.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

Register field	Entry
Authority	NWSC / Water authorities
Approval or compliance action	water and sewerage connection
Core file	connection documents, meters and discharge arrangements
Responsible owner	project manager during setup; compliance officer after launch
Board visibility	monthly until commissioning; quarterly after stabilization

Regulatory Sheet: NSSF and labor/OHS offices

Primary role: employee social security and workplace safety. Key documents/actions: registration, payroll compliance, safety records.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

Register field	Entry
Authority	NSSF and labor/OHS offices
Approval or compliance action	employee social security and workplace safety
Core file	registration, payroll compliance, safety records
Responsible owner	project manager during setup; compliance officer after launch
Board visibility	monthly until commissioning; quarterly after stabilization

Compliance Calendar

The compliance calendar should be reviewed monthly. Tax and payroll failures can quickly become investor issues even when production is performing well.

Frequency	Compliance action	Owner
Monthly	PAYE, NSSF, VAT returns where applicable, utility bills, safety toolbox records	Finance / HR / HSE
Quarterly	management accounts, board compliance report, supplier certificates	Finance / QC
Semi-annual	environmental housekeeping review, emergency drills	HSE / Plant manager
Annual	company filings, trade license renewal, insurance, product certification review	Compliance officer
As required	NEMA approval conditions, building inspections, standards audits	Project manager / QC

12. CAPEX and OPEX

The base case funding envelope is large enough to avoid a common manufacturing failure mode: buying the machine but underfunding power, yard, curing, fleet, lab and cash cycle.

TOTAL FUNDING

UGX 36.13 bn

including contingency and opening working capital

CAPEX SUBTOTAL

UGX 30.31 bn

before contingency and working capital

CONTINGENCY

8%

applied to CAPEX subtotal

The project requires a funding envelope that covers not just machinery, but site works, civil works, utilities, fleet, laboratory, pre-opening costs, contingency and working capital.

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CAPEX Schedule

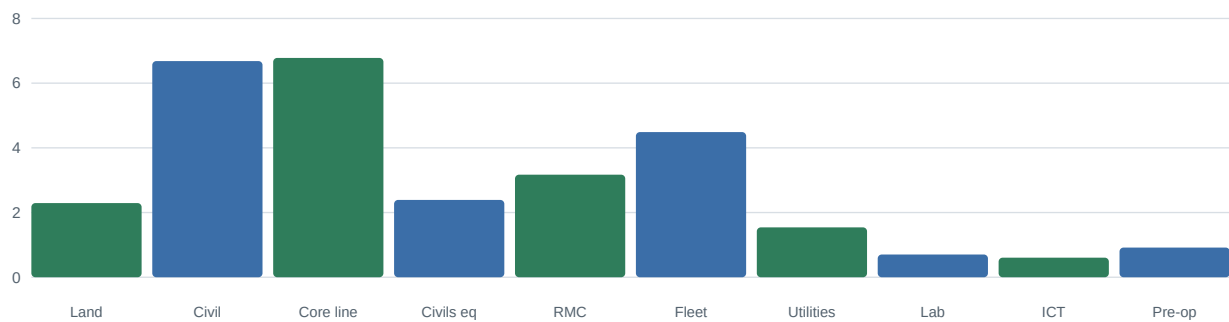
The schedule is a planning estimate. Supplier quotations and a local quantity surveyor should replace these figures before financing.

CAPEX item	Amount	Scope note
Land lease premium, surveys and site enabling	UGX 2,350,000,000	5-8 acre industrial plot, drainage, fencing and ground preparation
Civil works and factory buildings	UGX 6,850,000,000	production hall, curing sheds, stockyard pavement, admin and lab
Block and paver vibropress line	UGX 6,950,000,000	automatic batching, mixer, block machine, molds, stacker and pallets
Kerb, channel and culvert production equipment	UGX 2,450,000,000	forms, cage rollers, pipe molds, cranes and vibration systems
Ready-mix plant and concrete pumps	UGX 3,250,000,000	30-60 m3 per hour batching plant and dispatch controls
Mobile plant and logistics fleet	UGX 4,600,000,000	wheel loader, forklifts, tipper, flatbeds, transit mixers and pickup
Power, water and utilities infrastructure	UGX 1,580,000,000	transformer, wiring, borehole/NWSC interface, tanks and recycling
Laboratory and quality control equipment	UGX 720,000,000	cube press, molds, sieves, scales, slump kits and curing tanks
ERP, weighbridge, security and ICT	UGX 620,000,000	dispatch control, accounting, CCTV, access control and weighbridge
Pre-opening costs and training	UGX 940,000,000	technical commissioning, launch marketing, permits and operator training
Contingency	UGX 2,424,800,000	8% planning contingency
Opening working capital	UGX 3,400,000,000	cement, aggregate, payroll, spares and receivables buffer
Total funding envelope	UGX 36,134,800,000	base-case financing need

CAPEX Visual

Machinery, civil works and fleet dominate the investment. Underfunding the yard and utilities would directly reduce factory performance.

CAPEX by category, UGX bn



OPEX Model

OPEX discipline depends on three routines: batch yield, maintenance uptime and cash collection. Management should review each weekly.

OPEX category	Year 2027	Mature year	Management control
Cement and aggregates	largest variable cost	largest variable cost	supplier contracts, moisture control, waste reduction
Labor and payroll	UGX 2.15 bn	UGX 4.78 bn	shift planning and productivity per employee
Utilities	UGX 0.58 bn	UGX 1.64 bn	submetering, power factor and water recycling
Maintenance	UGX 0.42 bn	UGX 1.61 bn	preventive maintenance and spares planning
Sales/admin/logistics overhead	UGX 1.05 bn	UGX 3.14 bn	route economics and credit control

Cement Cost Sensitivity

A cement price increase flows directly into gross margin unless sales prices, mix efficiency or procurement terms adjust.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

Control	Practical action
Budget owner	assign named manager
Variance threshold	investigate exceptions above management tolerance
Frequency	weekly for production, monthly for board reporting
Evidence	ERP, batch tickets, stock counts and supplier invoices

Aggregate and Sand Cost Sensitivity

Delivered aggregate cost depends on quarry distance, truck utilization, moisture and rejection rates.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

Control	Practical action
Budget owner	assign named manager
Variance threshold	investigate exceptions above management tolerance
Frequency	weekly for production, monthly for board reporting
Evidence	ERP, batch tickets, stock counts and supplier invoices

Labor Productivity

Labor should be measured per saleable unit, not headcount alone. Automation only pays off when operators, maintenance and QC routines work together.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

Control	Practical action
Budget owner	assign named manager
Variance threshold	investigate exceptions above management tolerance
Frequency	weekly for production, monthly for board reporting
Evidence	ERP, batch tickets, stock counts and supplier invoices

Maintenance Budget

Preventive maintenance protects production days. The budget should include molds, wear parts, hydraulic components, vibration motors and critical sensors.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

Control	Practical action
Budget owner	assign named manager
Variance threshold	investigate exceptions above management tolerance
Frequency	weekly for production, monthly for board reporting
Evidence	ERP, batch tickets, stock counts and supplier invoices

Fleet and Distribution Cost

Freight must be priced by zone. A product can be profitable ex-works and loss-making after poor route planning.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

Control	Practical action
Budget owner	assign named manager
Variance threshold	investigate exceptions above management tolerance
Frequency	weekly for production, monthly for board reporting
Evidence	ERP, batch tickets, stock counts and supplier invoices

Working Capital

Receivables, cement stock, pallets and curing inventory can absorb cash faster than accounting profit suggests.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

Control	Practical action
Budget owner	assign named manager
Variance threshold	investigate exceptions above management tolerance
Frequency	weekly for production, monthly for board reporting
Evidence	ERP, batch tickets, stock counts and supplier invoices

Insurance and Security

The plant should carry property, machinery breakdown, vehicle, liability, workers and goods-in-transit cover.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

Control	Practical action
Budget owner	assign named manager
Variance threshold	investigate exceptions above management tolerance
Frequency	weekly for production, monthly for board reporting
Evidence	ERP, batch tickets, stock counts and supplier invoices

Management Controls

ERP, weighbridge, stock counts and invoice matching protect against shrinkage and unbilled deliveries.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

Control	Practical action
Budget owner	assign named manager
Variance threshold	investigate exceptions above management tolerance
Frequency	weekly for production, monthly for board reporting
Evidence	ERP, batch tickets, stock counts and supplier invoices

13. 10-Year Financial Forecast

This model is an investor screening case. It should be rebuilt in spreadsheet form with supplier quotes, financing terms, tax advice and signed customer pipeline before investment approval.

 FORECAST PERIOD

2026-2035

10-year horizon requested

 COMMERCIAL LAUNCH

2027

first full operating year

 CURRENCY

UGX only

all figures in nominal UGX

The 2026-2035 model shows a setup year followed by staged revenue growth, improving gross margin, disciplined OPEX and meaningful cash generation after ramp-up.

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Forecast Summary Table

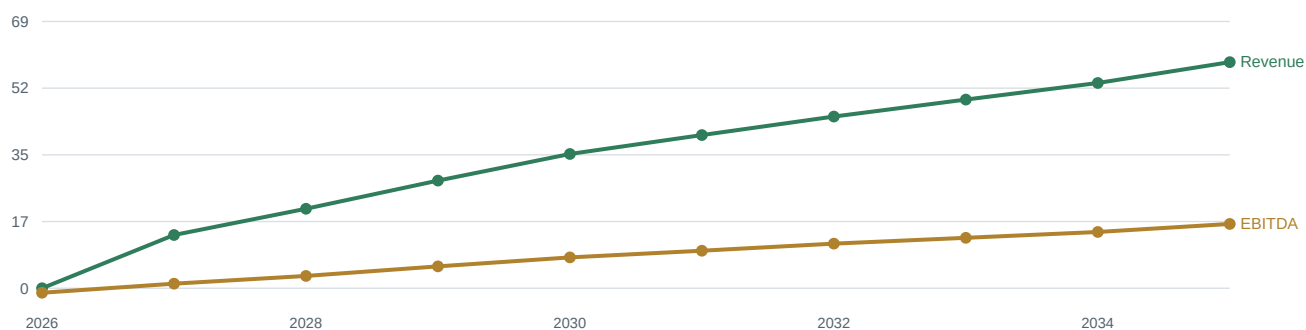
The unlevered free cash flow line treats 2026 as the investment year and excludes debt service. Financing structure will change shareholder returns and lender coverage metrics.

Year	Revenue	Gross profit	EBITDA	EBITDA margin	Tax	Unlevered FCF
2026	UGX 0.00 bn	UGX 0.00 bn	UGX -1.17 bn	0.0%	UGX 0.00 bn	UGX -37.30 bn
2027	UGX 13.80 bn	UGX 5.38 bn	UGX 1.18 bn	8.6%	UGX 0.00 bn	UGX 1.00 bn
2028	UGX 20.60 bn	UGX 8.45 bn	UGX 3.19 bn	15.5%	UGX 0.22 bn	UGX 2.75 bn
2029	UGX 27.90 bn	UGX 12.00 bn	UGX 5.67 bn	20.3%	UGX 0.97 bn	UGX 4.42 bn
2030	UGX 34.80 bn	UGX 15.31 bn	UGX 7.99 bn	23.0%	UGX 1.66 bn	UGX 5.98 bn
2031	UGX 39.70 bn	UGX 17.86 bn	UGX 9.72 bn	24.5%	UGX 2.18 bn	UGX 7.08 bn
2032	UGX 44.50 bn	UGX 20.47 bn	UGX 11.56 bn	26.0%	UGX 2.73 bn	UGX 8.25 bn
2033	UGX 48.90 bn	UGX 22.74 bn	UGX 13.07 bn	26.7%	UGX 3.19 bn	UGX 9.16 bn
2034	UGX 53.20 bn	UGX 25.00 bn	UGX 14.58 bn	27.4%	UGX 3.64 bn	UGX 10.06 bn
2035	UGX 58.60 bn	UGX 27.84 bn	UGX 16.66 bn	28.4%	UGX 4.26 bn	UGX 11.35 bn

Revenue and EBITDA Trend

The model depends on ramp-up. If capacity utilization grows slower than planned, management must protect cash by staging product launches and controlling credit.

Revenue and EBITDA, UGX bn



Revenue Mix, UGX bn

Blocks dominate early utilization; ready-mix and civil products increase as the factory builds contractor credibility and fleet capability.

Year	Blocks	Pavers	Civils	Ready-mix	Total
2026	0.00	0.00	0.00	0.00	0.00
2027	5.80	4.14	2.21	1.66	13.80
2028	8.65	6.18	3.30	2.47	20.60
2029	11.72	8.37	4.46	3.35	27.90
2030	12.18	9.74	6.26	6.61	34.80
2031	13.89	11.12	7.15	7.54	39.70
2032	15.57	12.46	8.01	8.46	44.50
2033	17.11	13.69	8.80	9.29	48.90
2034	18.62	14.90	9.58	10.11	53.20
2035	20.51	16.41	10.55	11.13	58.60

Financial Sensitivity

The most important investor sensitivities are utilization, cement cost, receivable days and product mix. These should become lender-case sensitivities in the next model version.

Scenario	Change	Impact	Management response
Base case	as modeled	positive NPV and mature EBITDA margin above 20%	execute staged launch
Slow ramp	revenue 15% lower in 2027-2029	payback moves later and working capital tightens	defer RMC fleet and preserve cash
Cement shock	cement cost 12% higher	gross margin pressure in blocks and pavers	price escalation, recipe efficiency and supplier split
Receivable stress	DSO +30 days	funding need increases materially	tighten credit and deposits
Premium upside	paver/civils mix +10%	margin improves and freight economics improve	market premium profiles and municipal contracts

Assumption: Utilization

The model assumes the core line ramps from single-shift launch to higher utilization by product family. Utilization is the main break-even driver.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

Model item	Base-case treatment
Currency	UGX only
Inflation	embedded in nominal revenue and cost ramp
Financing	unlevered forecast; debt structure not included in cash flow
Decision gate	replace assumptions with quotes and customer evidence

Assumption: Gross Margin

Gross margin improves as rejects reduce, procurement is contracted, premium pavers grow and ready-mix batching becomes disciplined.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

Model item	Base-case treatment
Currency	UGX only
Inflation	embedded in nominal revenue and cost ramp
Financing	unlevered forecast; debt structure not included in cash flow
Decision gate	replace assumptions with quotes and customer evidence

Assumption: Payroll

Payroll includes production, QC, maintenance, sales, dispatch, admin and security. Employer statutory costs should be confirmed.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

Model item	Base-case treatment
Currency	UGX only
Inflation	embedded in nominal revenue and cost ramp
Financing	unlevered forecast; debt structure not included in cash flow
Decision gate	replace assumptions with quotes and customer evidence

Assumption: Tax

A generic income tax assumption is included for modeling. Final tax treatment, depreciation and incentives require professional advice.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

Model item	Base-case treatment
Currency	UGX only
Inflation	embedded in nominal revenue and cost ramp
Financing	unlevered forecast; debt structure not included in cash flow
Decision gate	replace assumptions with quotes and customer evidence

Assumption: Maintenance CAPEX

Maintenance capital expenditure increases as plant utilization rises and molds, pallets and fleet require replacement.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

Model item	Base-case treatment
Currency	UGX only
Inflation	embedded in nominal revenue and cost ramp
Financing	unlevered forecast; debt structure not included in cash flow
Decision gate	replace assumptions with quotes and customer evidence

Assumption: Working Capital

Opening working capital covers raw material, payroll, spares, receivables and curing inventory during ramp-up.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

Model item	Base-case treatment
Currency	UGX only
Inflation	embedded in nominal revenue and cost ramp
Financing	unlevered forecast; debt structure not included in cash flow
Decision gate	replace assumptions with quotes and customer evidence

Debt Sizing Considerations

Lenders will focus on DSCR, collateral, sponsor equity, offtake quality, construction risk and working-capital controls.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

Model item	Base-case treatment
Currency	UGX only
Inflation	embedded in nominal revenue and cost ramp
Financing	unlevered forecast; debt structure not included in cash flow
Decision gate	replace assumptions with quotes and customer evidence

Equity Return Considerations

Equity upside depends on utilization, disciplined credit, product mix and ability to replicate distribution into regional markets.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

Model item	Base-case treatment
Currency	UGX only
Inflation	embedded in nominal revenue and cost ramp
Financing	unlevered forecast; debt structure not included in cash flow
Decision gate	replace assumptions with quotes and customer evidence

14. Risk Register

Risk ownership should be integrated into weekly management reporting during setup and monthly board reporting after launch.



RISKS TRACKED

15

strategic, operating, market and compliance risks



TOP SEVERITY

quality/cash

failed product or receivables can erode trust quickly



CONTROL STYLE

owner-led

each risk must have a named owner and metric

The project has manageable risks if they are owned early. The highest risks are raw material volatility, quality failures, utility reliability, credit control, environmental approvals and commissioning discipline.

1

Investor lens

What the chapter means for return, risk and bankability.

2

Operating lens

Factory routines, people, quality and throughput implications.

3

Decision lens

Actions required before financial close and procurement.

Risk Register Master Table

The risk register should be maintained as a live operating document. Severity should be recalibrated after site selection, supplier contracting and customer pipeline validation.

Risk	Likelihood	Impact	Mitigation
Cement price spike	High	High	Supplier framework agreements, monthly price index, dual sourcing and cement silo stock discipline.
Aggregate moisture variation	High	Medium	Moisture testing by shift, covered bays, calibration of water dosing and batch tickets.
Power interruption	Medium	High	UEDCL industrial connection, standby generator, maintenance windows and production buffer stock.
Weak curing discipline	Medium	High	Curing chambers, QC sign-off before dispatch and batch traceability by product lot.
Price undercutting by informal block makers	High	Medium	Segment into certified quality, delivery reliability, credit controls and contractor education.
Slow receivables from contractors	High	High	Credit approval, milestone billing, retention caps and early-payment discounts.
Environmental approval delays	Medium	High	Early NEMA screening, project brief/ESIA route, wetland setbacks and washout management.
Equipment commissioning delay	Medium	High	Supplier FAT, local spares pack, liquidated damages and operator training before shipment.
RMC slump or strength disputes	Medium	High	Mix design approvals, cube records, delivery tickets, slump testing and site acceptance protocol.
Fleet utilization below plan	Medium	Medium	Dispatch optimization, route costing, backhaul planning and outsourced peak capacity.
Foreign exchange exposure on machinery	Medium	Medium	Lock purchase timing, hedge deposits where available, maintain UGX funding contingency.
Regulatory non-compliance	Low	High	Compliance calendar covering URSB, URA, NSSF, NEMA, UNBS, city permits and OHS.
Safety incident	Medium	High	Traffic separation, LOTO, PPE, lifting plans, near-miss reporting and supervisor accountability.
Water shortage or abstraction limit	Medium	Medium	NWSC/borehole redundancy, rainwater harvesting, recycling and water meter dashboard.
Expansion execution stretch	Medium	Medium	Stage-gate East African roll-out; do not launch depots before service metrics stabilize.

Risk Detail: Cement price spike

Likelihood: High. Impact: High. Mitigation: Supplier framework agreements, monthly price index, dual sourcing and cement silo stock discipline.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

Field	Entry
Risk event	Cement price spike
Primary owner	to be assigned during project setup
Early warning	define measurable trigger before commissioning
Control evidence	policy, records, dashboard or inspection proof
Board escalation	required when likelihood or impact moves to high

Risk Detail: Aggregate moisture variation

Likelihood: High. Impact: Medium. Mitigation: Moisture testing by shift, covered bays, calibration of water dosing and batch tickets.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

Field	Entry
Risk event	Aggregate moisture variation
Primary owner	to be assigned during project setup
Early warning	define measurable trigger before commissioning
Control evidence	policy, records, dashboard or inspection proof
Board escalation	required when likelihood or impact moves to high

Risk Detail: Power interruption

Likelihood: Medium. Impact: High. Mitigation: UEDCL industrial connection, standby generator, maintenance windows and production buffer stock.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

Field	Entry
Risk event	Power interruption
Primary owner	to be assigned during project setup
Early warning	define measurable trigger before commissioning
Control evidence	policy, records, dashboard or inspection proof
Board escalation	required when likelihood or impact moves to high

Risk Detail: Weak curing discipline

Likelihood: Medium. Impact: High. Mitigation: Curing chambers, QC sign-off before dispatch and batch traceability by product lot.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

Field	Entry
Risk event	Weak curing discipline
Primary owner	to be assigned during project setup
Early warning	define measurable trigger before commissioning
Control evidence	policy, records, dashboard or inspection proof
Board escalation	required when likelihood or impact moves to high

Risk Detail: Price undercutting by informal block makers

Likelihood: High. Impact: Medium. Mitigation: Segment into certified quality, delivery reliability, credit controls and contractor education.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

Field	Entry
Risk event	Price undercutting by informal block makers
Primary owner	to be assigned during project setup
Early warning	define measurable trigger before commissioning
Control evidence	policy, records, dashboard or inspection proof
Board escalation	required when likelihood or impact moves to high

Risk Detail: Slow receivables from contractors

Likelihood: High. Impact: High. Mitigation: Credit approval, milestone billing, retention caps and early-payment discounts.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

Field	Entry
Risk event	Slow receivables from contractors
Primary owner	to be assigned during project setup
Early warning	define measurable trigger before commissioning
Control evidence	policy, records, dashboard or inspection proof
Board escalation	required when likelihood or impact moves to high

Risk Detail: Environmental approval delays

Likelihood: Medium. Impact: High. Mitigation: Early NEMA screening, project brief/ESIA route, wetland setbacks and washout management.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

Field	Entry
Risk event	Environmental approval delays
Primary owner	to be assigned during project setup
Early warning	define measurable trigger before commissioning
Control evidence	policy, records, dashboard or inspection proof
Board escalation	required when likelihood or impact moves to high

Risk Detail: Equipment commissioning delay

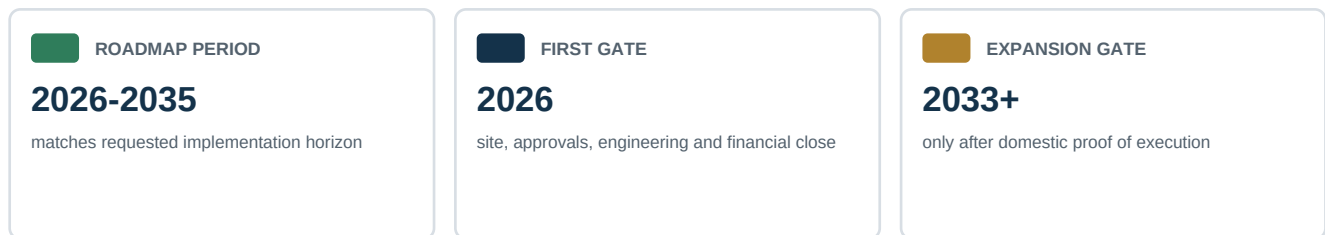
Likelihood: Medium. Impact: High. Mitigation: Supplier FAT, local spares pack, liquidated damages and operator training before shipment.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

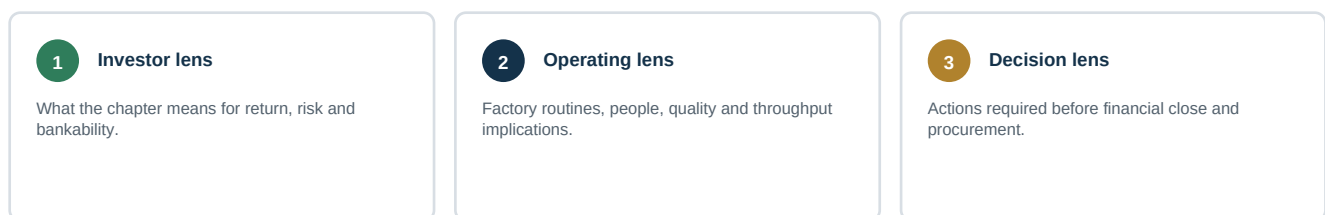
Field	Entry
Risk event	Equipment commissioning delay
Primary owner	to be assigned during project setup
Early warning	define measurable trigger before commissioning
Control evidence	policy, records, dashboard or inspection proof
Board escalation	required when likelihood or impact moves to high

15. Implementation Roadmap 2026-2035

Each year has a gating logic. The factory should not accelerate expansion if quality, cash conversion or service levels are below thresholds.



The roadmap is staged to reduce execution risk: approvals and procurement in 2026, core production in 2027, product expansion through 2030, operating maturity through 2032 and regional options from 2033.



Roadmap Overview

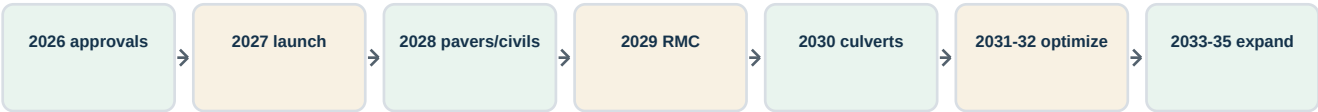
The roadmap turns a single factory into a disciplined operating platform. The sponsor should approve each stage only when evidence from the prior stage is acceptable.

Year	Milestone
2026	Financial close, site acquisition, ESIA route, detailed engineering, procurement shortlist and anchor customer MOUs.
2027	Commission block/paver line, launch Mbarara sales office, establish QC lab and secure first municipal/institutional contracts.
2028	Add kerbstone/channel molds, open contractor credit desk and deepen Bushenyi-Ntungamo-Kabale distribution.
2029	Commission ready-mix plant, deploy transit mixers and target commercial slabs, hospitals, schools and estates.
2030	Introduce culvert line, strengthen road-infrastructure bid capability and formalize Rwanda/DRC route intelligence.
2031	Launch certified permeable paver range, improve water recycling and target climate-resilient urban drainage projects.
2032	Automate pallet handling, expand stockyard, introduce second shift for pavers and standardize dealer scorecards.
2033	Open first cross-border depot or bonded distribution partner depending on route economics and receivable quality.
2034	Evaluate second production cell, larger RMC batch capacity and strategic alliance with quarry or aggregate supplier.
2035	Regional consolidation, product certification refresh, refinance or exit-readiness review and ESG impact reporting.

Roadmap Visual

The staged plan protects capital by linking new product launches and regional moves to proven factory controls.

Implementation stage gates



Roadmap Year: 2026

Financial close, site acquisition, ESIA route, detailed engineering, procurement shortlist and anchor customer MOUs.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

Workstream	Focus
Commercial	pipeline, pricing, customer evidence and collections
Operations	throughput, quality, maintenance and safety
Finance	budget, working capital and variance control
Compliance	approvals, renewals and permit conditions
People	hiring, training and accountability

Roadmap Year: 2027

Commission block/paver line, launch Mbarara sales office, establish QC lab and secure first municipal/institutional contracts.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

Workstream	Focus
Commercial	pipeline, pricing, customer evidence and collections
Operations	throughput, quality, maintenance and safety
Finance	budget, working capital and variance control
Compliance	approvals, renewals and permit conditions
People	hiring, training and accountability

Roadmap Year: 2028

Add kerbstone/channel molds, open contractor credit desk and deepen Bushenyi-Ntungamo-Kabale distribution.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

Workstream	Focus
Commercial	pipeline, pricing, customer evidence and collections
Operations	throughput, quality, maintenance and safety
Finance	budget, working capital and variance control
Compliance	approvals, renewals and permit conditions
People	hiring, training and accountability

Roadmap Year: 2029

Commission ready-mix plant, deploy transit mixers and target commercial slabs, hospitals, schools and estates.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

Workstream	Focus
Commercial	pipeline, pricing, customer evidence and collections
Operations	throughput, quality, maintenance and safety
Finance	budget, working capital and variance control
Compliance	approvals, renewals and permit conditions
People	hiring, training and accountability

Roadmap Year: 2030

Introduce culvert line, strengthen road-infrastructure bid capability and formalize Rwanda/DRC route intelligence.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

Workstream	Focus
Commercial	pipeline, pricing, customer evidence and collections
Operations	throughput, quality, maintenance and safety
Finance	budget, working capital and variance control
Compliance	approvals, renewals and permit conditions
People	hiring, training and accountability

Roadmap Year: 2031

Launch certified permeable paver range, improve water recycling and target climate-resilient urban drainage projects.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

Workstream	Focus
Commercial	pipeline, pricing, customer evidence and collections
Operations	throughput, quality, maintenance and safety
Finance	budget, working capital and variance control
Compliance	approvals, renewals and permit conditions
People	hiring, training and accountability

Roadmap Year: 2032

Automate pallet handling, expand stockyard, introduce second shift for pavers and standardize dealer scorecards.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

Workstream	Focus
Commercial	pipeline, pricing, customer evidence and collections
Operations	throughput, quality, maintenance and safety
Finance	budget, working capital and variance control
Compliance	approvals, renewals and permit conditions
People	hiring, training and accountability

Roadmap Year: 2033

Open first cross-border depot or bonded distribution partner depending on route economics and receivable quality.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

Workstream	Focus
Commercial	pipeline, pricing, customer evidence and collections
Operations	throughput, quality, maintenance and safety
Finance	budget, working capital and variance control
Compliance	approvals, renewals and permit conditions
People	hiring, training and accountability

Roadmap Year: 2034

Evaluate second production cell, larger RMC batch capacity and strategic alliance with quarry or aggregate supplier.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

Workstream	Focus
Commercial	pipeline, pricing, customer evidence and collections
Operations	throughput, quality, maintenance and safety
Finance	budget, working capital and variance control
Compliance	approvals, renewals and permit conditions
People	hiring, training and accountability

Roadmap Year: 2035

Regional consolidation, product certification refresh, refinance or exit-readiness review and ESG impact reporting.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

Workstream	Focus
Commercial	pipeline, pricing, customer evidence and collections
Operations	throughput, quality, maintenance and safety
Finance	budget, working capital and variance control
Compliance	approvals, renewals and permit conditions
People	hiring, training and accountability

16. East African Expansion Plan

Potential expansion routes include western Uganda, Rwanda-facing routes, DRC-facing routes, northern Tanzania and selected institutional customers. The plan must remain UGX-based for this report.

EXPANSION MODE

depot first

avoid premature second factory

PRIORITY ROUTE

Rwanda/DRC corridors

after domestic distribution maturity

GATE METRIC

cash conversion

do not export receivables problems

Regional expansion should be distribution-led before it is manufacturing-led. The factory should first prove delivery economics, payment discipline and product acceptance across corridors.

1 Investor lens

What the chapter means for return, risk and bankability.

2 Operating lens

Factory routines, people, quality and throughput implications.

3 Decision lens

Actions required before financial close and procurement.

Regional Expansion Logic

Concrete products are freight-sensitive. Expansion should prioritize higher-value products, reliable dealer partners and pre-sold project loads before standard blocks travel long distances.

Market	Entry logic	Best-fit products	Gate condition
Western Uganda deepening	direct fleet and dealers	blocks, pavers, kerbs	service level above target
Rwanda-facing corridor	partner depot after route tests	pavers, kerbs, premium products	repeat orders and low damage
DRC-facing corridor	project-led export trials	culverts, channels, pavers	secured payment and border logistics
Northern Tanzania	opportunistic projects	pavers and RMC not favored except local projects	route economics proven
Second plant option	only after 2034 review	full portfolio	capacity constraint and positive NPV

Export Readiness Gate

Confirm product certification, packaging, palletization, border documents, customer payment terms, insurance and route damage rates.

Expansion should be stopped or slowed if product damage, border delays or receivable days exceed thresholds.

Gate	Evidence
Demand	repeat orders or signed project pipeline
Route economics	delivered margin after freight and damage
Cash	advance payment, secure credit or insured receivable
Operations	no degradation in domestic service levels

Depot Model

Start with a leased stock point and local sales partner rather than building a new plant. Track stock turns, damage and receivable days.

Expansion should be stopped or slowed if product damage, border delays or receivable days exceed thresholds.

Gate	Evidence
Demand	repeat orders or signed project pipeline
Route economics	delivered margin after freight and damage
Cash	advance payment, secure credit or insured receivable
Operations	no degradation in domestic service levels

Rwanda Corridor

Favor premium pavers, kerbs and packaged products where freight can be justified. Avoid low-margin block exports unless project-specific.

Expansion should be stopped or slowed if product damage, border delays or receivable days exceed thresholds.

Gate	Evidence
Demand	repeat orders or signed project pipeline
Route economics	delivered margin after freight and damage
Cash	advance payment, secure credit or insured receivable
Operations	no degradation in domestic service levels

DRC Corridor

Use project-led shipments with advance payment or secured terms. Culverts and channels may fit road/drainage demand when payment risk is controlled.

Expansion should be stopped or slowed if product damage, border delays or receivable days exceed thresholds.

Gate	Evidence
Demand	repeat orders or signed project pipeline
Route economics	delivered margin after freight and damage
Cash	advance payment, secure credit or insured receivable
Operations	no degradation in domestic service levels

Tanzania Link

Treat as opportunistic until a local partner proves demand and route economics.

Expansion should be stopped or slowed if product damage, border delays or receivable days exceed thresholds.

Gate	Evidence
Demand	repeat orders or signed project pipeline
Route economics	delivered margin after freight and damage
Cash	advance payment, secure credit or insured receivable
Operations	no degradation in domestic service levels

Regional Brand System

Maintain the same batch traceability, QC sheets and delivery documentation across all markets.

Expansion should be stopped or slowed if product damage, border delays or receivable days exceed thresholds.

Gate	Evidence
Demand	repeat orders or signed project pipeline
Route economics	delivered margin after freight and damage
Cash	advance payment, secure credit or insured receivable
Operations	no degradation in domestic service levels

Second Factory Decision

A second factory should require proven demand, land availability, management depth, positive NPV and a clear raw-material advantage.

Expansion should be stopped or slowed if product damage, border delays or receivable days exceed thresholds.

Gate	Evidence
Demand	repeat orders or signed project pipeline
Route economics	delivered margin after freight and damage
Cash	advance payment, secure credit or insured receivable
Operations	no degradation in domestic service levels

17. Appendices

Appendix material is deliberately practical: assumptions, scorecards, registers, quality forms, due diligence questions and source references.



APPENDIX TYPE

templates

ready to convert into operational controls



EVIDENCE

sources

public-source reference list included



USE

diligence

board, lender and management follow-up

The appendices provide detailed worksheets and templates that management can convert into live operating documents.

1

Investor lens

What the chapter means for return, risk and bankability.

2

Operating lens

Factory routines, people, quality and throughput implications.

3

Decision lens

Actions required before financial close and procurement.

Appendix A: Assumption Register

The assumption register should be transferred into the financial model and marked as assumption, quote, contract, policy or actual result.

Assumption	Base case	Owner	Validation method
Funding envelope	UGX 36.13 bn	Sponsor/CFO	supplier quotes and QS estimate
Commercial launch	2027	Project manager	commissioning schedule
Gross margin maturity	47.5%	Plant manager/CFO	product-level costing
Working capital	UGX 3.40 bn	CFO	receivables and inventory schedule
Water demand	45-125 m3/day	Operations	metering and engineering design
Power demand	450-750 kVA	Electrical engineer	load schedule and utility confirmation

Appendix B: Supplier Scorecard

Supplier management should be evidence-based. The scorecard should be reviewed monthly for cement, sand, stone dust, aggregate, fuel, steel, packaging and admixtures.

Criteria	Weight	Evidence
Delivered price stability	25%	invoice trend and escalation notices
On-time delivery	20%	delivery logs
Quality pass rate	25%	receipt inspection and lab results
Documentation	10%	certificates, tax invoices, batch details
Commercial terms	10%	credit, minimum order, payment flexibility
Responsiveness	10%	issue resolution time

Appendix C: Customer Credit Scorecard

Credit should be earned, not granted by relationship pressure. The factory should separate sales approval from credit approval.

Criteria	Score rule
Customer identity	registered business, verified directors and site contact
Project evidence	BOQ, contract, owner and payment source
Payment history	cash first, then small limit, then review
Security	purchase order, deposit, guarantee or retention cap
Dispute history	delivery sign-off and complaint behavior
Limit review	monthly aging and exposure report

Appendix D: Daily Production Report

A simple daily report protects the investor from hidden underperformance. It should be signed by production, QC and dispatch.

Field	Entry
Date / shift	
Product and mold	
Planned output	
Actual green output	
Rejected units	
Curing lot reference	
Downtime minutes and cause	
Material variance	
QC release notes	

Appendix E: QC Release Form

No finished product should move from curing to saleable stock without a release record.

Check	Pass / fail / notes
Product and lot number	
Curing start and release date	
Visual defects	
Dimensional tolerance	
Cube test reference	
Breakage during handling	
Release decision	
QC signature	

Appendix F: RMC Delivery Ticket Fields

Ready-mix disputes are best prevented at dispatch. The delivery ticket should align with batch records and site acceptance.

Field	Purpose
Customer and site	confirm delivery responsibility
Concrete grade and slump	match order and site requirement
Batch time and truck departure	control time-sensitive delivery
Water added at plant	trace water-cement discipline
Water added on site	allowed only by approved protocol
Slump result	site acceptance evidence
Customer sign-off	billing and dispute control

Appendix G: Board KPI Pack

The board should see a concise operational dashboard, not only accounting profit. The dashboard should connect production, quality, sales, cash and risk.

KPI	Target direction	Why it matters
Revenue by product family	up with mix discipline	growth and portfolio health
Gross margin by product	stable or improving	pricing and raw material control
Reject rate	down	quality and cost leakage
On-time delivery	up	customer trust
Receivable days	down	cash conversion
Inventory age	controlled	curing discipline and stock rotation
Lost-time incidents	zero	safety and operational continuity

Appendix H: Site Due Diligence Checklist

Land that appears cheap can become expensive if access, drainage, utilities, zoning or ground conditions are poor.

Item	Required evidence
Title / lease	lawyer-reviewed tenure and encumbrance search
Zoning	industrial use compatibility and city planning confirmation
Access	truck route, turning radius and road condition
Geotechnical	bearing capacity, groundwater and earthworks requirement
Drainage	stormwater outfall and flood risk
Utilities	power route, water source and communications
Neighbors	dust, noise, operating hours and social license

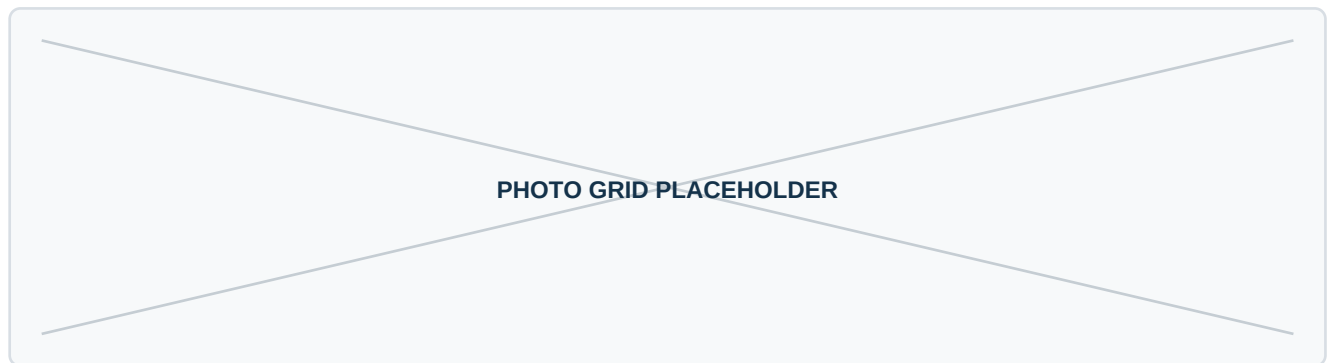
Appendix I: ESG and Community Commitments

A practical ESG plan should focus on dust, noise, water, washout, road safety, jobs, skills, local sourcing and transparent complaint handling.

Area	Commitment
Dust	covered stockpiles, water sprays and housekeeping
Water	metering, recycling and compliant discharge
Waste	rework broken products where possible; manage cement washout
Safety	traffic separation, PPE, training and incident reporting
Jobs	local hiring and structured operator training
Community	complaint log and response time standard

Appendix J: Product Photo Placeholders

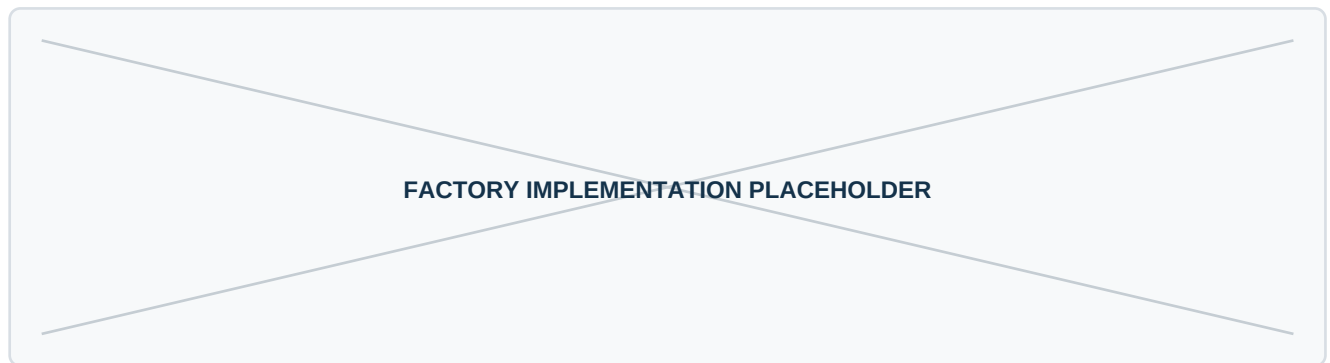
The final investor deck or lender appendix should insert product photos from trial production, not generic stock imagery. Photos should show dimensions, surface finish, packaging, curing and loading.



Insert four to six verified product photos after mold trials.

Appendix K: Factory Photo Placeholders

During implementation, replace this placeholder with site photos, layout progress, equipment installation and commissioning evidence.



Insert site, equipment, lab and yard photographs.

Appendix L: Source List

The report uses public sources for context and internally generated assumptions for the financial model. Source links should be refreshed before financing.

Source	URL	Use in report
World Bank Uganda overview	https://www.worldbank.org/ext/en/country/uganda	Macro growth, population, industrialization context
UBOS 2024 Census portal	https://statistics.ubos.org/nphc/index	Population and household evidence base
UBOS publications explorer	https://www.ubos.org/?p_id=126&pagename=explore-publications	Annual GDP, construction input price index and national accounts publications
NEMA ESIA process	https://www.nema.go.ug/en/esia/	Environmental assessment route for project brief, ToR or ESIA
URSB business registration	https://ursb.go.ug/services/business-registration/	Company registration mandate and business formalization benefits
UIA investment FAQs	https://ugandainvest.go.ug/contact/faqs/	Investment licensing and investor facilitation context
UIA industrial parks	https://ugandainvest.go.ug/parks/	Industrial park infrastructure and land facilitation context
ERA UEDCL transition	https://www.era.go.ug/event/uedcl-take-over/	Electricity distribution transition to UEDCL from April 2025
NWSC new connections	https://www.nwsc.co.ug/new-connections/	Water connection process and supporting documents
NBRB concrete guidance	https://nbrb.go.ug/wp-content/uploads/2024/09/GUIDELINES-FOR-TH E-DESIGN-AND-CONSTRUCTION-OF-STRUCTURES-WITH-STEEL-CONCRETE-AND-TIMBER.pdf	Concrete water-cement, cube testing, curing and compaction guidance
UNBS paver standard webstore	https://webstore.unbs.go.ug/store.php?preview=&src=401	Precast concrete paving block standard reference
Nicomah Uganda	https://nicomah.com/	Western Uganda construction material and concrete product supplier reference
Master Industries Uganda	https://masterindustriesug.com/	Concrete blocks, pavers, kerbs and culvert competitor reference
Tororo Cement	https://www.tororocement.com/	Cement producer and construction material product reference
Hima Cement	https://himacement.co/	Cement products, blocks and precast reference
UIA cement factory news	https://ugandainvest.go.ug/new-300-million-dollar-cement-factory-opening-in-uganda/	Uganda cement supply landscape context

Appendix Worksheet: Demand interviews

This worksheet should be completed during the next diligence phase for demand interviews.

Convert the findings into action items, model changes, contract conditions or board approvals.

Question	Evidence / answer
What decision depends on this workstream?	
Who owns the workstream?	
What evidence has been collected?	
What remains uncertain?	
What model assumption changes?	
What is the recommended board action?	

Appendix Worksheet: Supplier quotations

This worksheet should be completed during the next diligence phase for supplier quotations.

Convert the findings into action items, model changes, contract conditions or board approvals.

Question	Evidence / answer
What decision depends on this workstream?	
Who owns the workstream?	
What evidence has been collected?	
What remains uncertain?	
What model assumption changes?	
What is the recommended board action?	

Appendix Worksheet: Equipment factory visit

This worksheet should be completed during the next diligence phase for equipment factory visit.

Convert the findings into action items, model changes, contract conditions or board approvals.

Question	Evidence / answer
What decision depends on this workstream?	
Who owns the workstream?	
What evidence has been collected?	
What remains uncertain?	
What model assumption changes?	
What is the recommended board action?	

Appendix Worksheet: Civil works BoQ

This worksheet should be completed during the next diligence phase for civil works boq.

Convert the findings into action items, model changes, contract conditions or board approvals.

Question	Evidence / answer
What decision depends on this workstream?	
Who owns the workstream?	
What evidence has been collected?	
What remains uncertain?	
What model assumption changes?	
What is the recommended board action?	

Appendix Worksheet: Tax and incentive advice

This worksheet should be completed during the next diligence phase for tax and incentive advice.

Convert the findings into action items, model changes, contract conditions or board approvals.

Question	Evidence / answer
What decision depends on this workstream?	
Who owns the workstream?	
What evidence has been collected?	
What remains uncertain?	
What model assumption changes?	
What is the recommended board action?	

Appendix Worksheet: Environmental route confirmation

This worksheet should be completed during the next diligence phase for environmental route confirmation.

Convert the findings into action items, model changes, contract conditions or board approvals.

Question	Evidence / answer
What decision depends on this workstream?	
Who owns the workstream?	
What evidence has been collected?	
What remains uncertain?	
What model assumption changes?	
What is the recommended board action?	